

1 When do single-cell response distributions improve drug 2 retrieval?

4 Abstract

5 Artificial intelligence virtual cells aim to predict cellular responses to perturbations and, ultimately,
6 to guide intervention selection. Conventional transcriptomic drug retrieval represents each pertur-
7 bation by a single expression signature, whereas single-cell perturbation atlases preserve cell-to-cell
8 variation in perturbation responses. Here we define population-to-population drug retrieval (PopRe-
9 trieve), in which candidate drugs are ranked by comparing single-cell response populations, and ask
10 when this additional information improves drug ranking. We find that most of the gain over direction-
11 only mean matching is recovered by a mean-based score that also retains response magnitude, with
12 a smaller additional gain from retaining the full response population. The advantage of population-
13 level retrieval weakens as evaluation moves away from response matching: it disappears or reverses
14 under mechanism-of-action recovery, and most of its association with an external functional readout is
15 reproduced by response magnitude alone. The same held-out protein measurements favour different
16 retrieval strategies depending on whether they are summarized as means or retained as populations,
17 showing that the statistical form of an external evaluator can influence which retrieval strategy ap-
18 pears preferable. With candidate responses observed, population-level retrieval corrects an incorrect
19 top-ranked choice about four times as often as it overturns a correct one relative to magnitude-aware
20 mean matching, and about five times relative to direction-only matching, but the gain is confined to
21 the roughly one-fifth of queries for which mean matching leaves room for improvement. When candi-
22 date responses are predicted rather than observed, this oracle advantage disappears and can reverse
23 sign even when mean-level retrieval performance is preserved. Together, these results show that
24 the value of single-cell population information depends on whether information beyond the mean
25 changes candidate ordering and, when required, survives forward prediction.
26

27 Keywords: artificial intelligence virtual cell; single-cell perturbation; drug retrieval; population-to-
28 population retrieval; cellular heterogeneity; objective–utility mismatch

30 Introduction

31 Artificial intelligence virtual cells^{1,2} aim to predict cellular responses to perturbations^{3–5} and, ultimately,
32 to help prioritize interventions for testing. These goals pose two related problems. Forward prediction
33 asks what response a perturbation will produce^{6–9}; retrieval asks which perturbation best matches a
34 cellular response of interest¹⁰. Transcriptomic drug retrieval has traditionally represented each drug by
35 a single perturbation signature and ranked candidates by similarity or reversal^{11,12}. This approach has
36 made large drug libraries searchable, but collapses variation across cells into a single summary. Single-
37 cell perturbation atlases now capture this variation^{13–15}, including state-dependent responses¹⁶ and ef-
38 fects confined to minority cell populations¹⁷. Here we define population-to-population drug retrieval
39 (PopRetrieve), in which candidate drugs are ranked by comparing single-cell response populations, and
40 ask when this additional information improves drug ranking.

More detailed response information does not necessarily lead to a better drug ranking. Two drugs can have similar average responses yet differ in how individual cells respond, creating information that a population-aware method can use. But an apparent advantage over classical signature matching can arise for more than one reason. Population comparison retains response magnitude that directional similarity discards, so a gain over direction-only matching need not arise from within-population structure. The apparent benefit also depends on how retrieval quality is evaluated: a criterion that preserves full response distributions^{18,19} will naturally favour information that a population-aware scorer retains²⁰⁻²². Such gains establish better response matching, but not necessarily a better biological or functional ranking²³⁻²⁷.

We therefore decompose the problem into information retained, information that changes candidate ordering, and information that survives prediction. We first distinguish three scoring rules by how much of the response each retains: direction, direction plus magnitude, and the full response population. This separates gains recovered by restoring response magnitude from those that require information beyond the mean. We then evaluate the resulting rankings using increasingly task-distal criteria: response matching, mechanism recovery and external functional readouts^{28,29}. We use state-dependent perturbation responses as a concrete mechanism through which population information can affect retrieval. For such information to matter, it must be present and recoverable, change candidate ordering beyond what the mean already determines and, when candidate responses are not observed, survive forward prediction. This separates the existence of single-cell response structure from its value for intervention selection.

Here we show that most of the advantage of population comparison over direction-only signature matching is recovered by a mean-based score that also retains response magnitude, leaving a smaller gain that requires the full response population. The advantage of population comparison weakens as evaluation moves away from response matching: it disappears or reverses under mechanism-of-action recovery, and most of its association with an external functional readout is reproduced by response magnitude alone. The same held-out protein measurements favour different retrieval strategies depending only on whether they are summarized as means or retained as populations, showing that external evaluation is not automatically neutral. In a source-to-target intervention-identification task, population information provides a smaller additional gain beyond magnitude-aware mean matching when candidate responses are observed, and its whole advantage over direction-only mean matching is confined to the queries for which that baseline leaves room for improvement. That advantage does not survive when candidate responses must instead be predicted. Together, these results show that the value of single-cell population information depends not simply on whether additional cellular structure exists, but on whether it changes candidate ordering and survives to the final retrieval decision.

Results

Population-to-population retrieval generalizes classical signature matching

Population-to-population retrieval ranks candidates by comparing query and candidate response populations directly (Fig. 1a,b), whereas classical mean-signature retrieval^{11,12} first collapses each population to a single mean differential-expression vector. Retaining the cells makes response structure available that averaging removes, including differences between cellular states or subpopulations. The retrieval problem is unchanged; what changes is how much of the measured response reaches the ranking.

Two candidate populations can share the same mean response and still differ markedly in how individual cells respond (Fig. 1c,d). A population-aware score separates such candidates where a score

84 computed from the mean cannot, because any such score reads only the mean the two candidates share.
85 We therefore hold the candidate rankings fixed and judge them under criteria that progressively reduce
86 their coupling to the retrieval score: response matching, mechanism recovery and external functional
87 readouts (Fig. 1e,f).

88 Mean-signature and population-level retrieval are one task at two resolutions, connected by the
89 amount of within-population variation retained. As that variation vanishes, a response population col-
90 lapses onto its mean, and the energy distance between two such populations differs from the Euclidean
91 distance between their means only by a factor of two, which cannot reorder candidates. The magnitude-
92 aware mean L2 is therefore the zero-variance limit of energy-based population retrieval (Fig. 1g; Meth-
93 ods). The cosine used by classical connectivity scoring discards the length of that mean difference and
94 coarsens the same representation one step further (Fig. 1a,b). The implemented CMap-style cosine ap-
95 plies that same rule to the same mean vectors and is therefore identical to mean-cosine retrieval, whereas
96 the canonical L1000 weighted connectivity score remains a distinct rank-based score (Fig. 2f).

97 The coverage family provides a second continuum: sharpening its aggregation moves the score from
98 averaging subpopulation distances toward the worst-matched state, and a single undivided component
99 recovers the global distance (Fig. 1h; Methods). These relationships place classical signature matching
100 and population comparison on one axis of sensitivity to within-population structure (Supplementary
101 Table 1).

102 **Most of the gain over directional matching does not require the full response population**

103 Population-level retrieval substantially outperformed direction-only mean matching, and most of that
104 gain did not require the full response population. With candidate responses directly observed, so that
105 the full cellular populations are available to both approaches, energy-distance retrieval reached a Hit@1
106 of 0.836, macro-averaged across the seven task-setting combinations analysed here, against 0.388 for
107 mean-cosine retrieval and the numerically identical CMap-style cosine baseline^{11,12}. Between them sits
108 a magnitude-aware mean L2 at 0.788, applied to the same mean signatures: the step to that score is
109 +0.399 of the gap and the step from it to the full distribution a further +0.048 (Fig. 2a). Because that
110 first step changes both what the score reads, from direction alone to direction and magnitude, and the
111 geometry it uses, from angular similarity to Euclidean distance, mean L2 is a magnitude-aware mean
112 baseline rather than a pure magnitude ablation. The same ordering held in PCA space (Fig. 2a).

113 This ordering was not universal. On Frangieh Perturb-CITE-seq³⁰, the only one of the seven com-
114 binations not built from SciPlex3, the magnitude-aware mean L2 was the best of the nine scores (Hit@1
115 0.611), ahead of mean cosine (0.600) and of energy distance (0.578; Fig. 2b). There, retaining magnitude
116 helped and retaining the full response distribution did not.

117 Population-level scoring also improved response matching under partial observation, where only
118 part of each candidate population is available. The same rankings were judged by a regret criterion that
119 penalizes selecting a candidate whose per-cell outcomes are worse than the best available (Methods).
120 Across 765 queries, the worst-case subpopulation-coverage score reduced median regret by +0.031 rel-
121 ative to direction-only mean cosine (95% bootstrap CI +0.017 to +0.049), with lower regret on 56% of
122 queries (Fig. 2c).

123 Reducing the minority response state from half of the query population to a tenth did not erode the
124 energy-distance advantage consistently: it collapsed in one cell line and held in the other two (Fig. 2d).
125 The pattern belongs to the population-aware family rather than to one score. All five variants improved
126 on direction-only mean cosine, whereas against the magnitude-aware mean L2 the three distributional

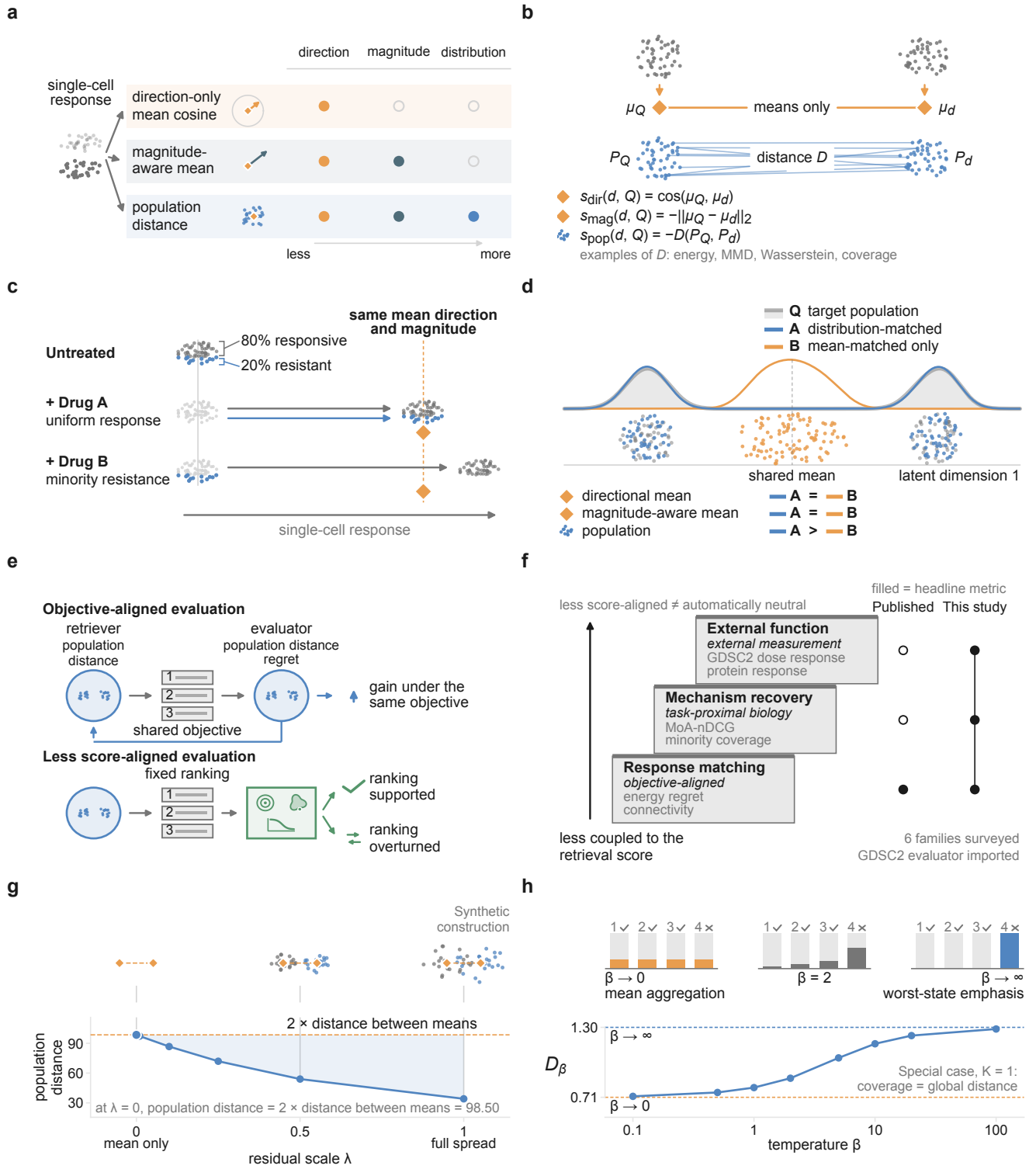


Figure 1: Perturbation responses can be represented and evaluated at multiple information resolutions. a–f, schematic; g, a seeded synthetic construction; h, an analytic illustration. a, Three scoring rules read one single-cell response population (grey, two states). Glyphs, the geometry each rule uses: a mean vector μ normalised to a unit circle, the same vector at its true length, and the population P with μ still on it (orange diamond, mean; blue cells, population). Matrix, the properties each rule retains (filled) or discards (open); the columns name properties and carry no quantity. **b**, One pair of populations and the three scoring rules: the mean route keeps one vector each, the population route compares the cells. **c**, Under drug B a resistant fifth does not move, yet the two treated means coincide in direction and in magnitude; faint cells, untreated. **d**, Three populations sharing one sample mean exactly, so both mean scores tie by construction while the population score, an energy distance over the drawn cells, prefers A. **e**, Objective-aligned, the score that produced the ranking also grades it; less score-aligned, the same ranking is judged by mechanism of action, target engagement or GDSC2 viability (icons). **f**, Three evaluation levels ordered by coupling to the retrieval score; filled, that level is a study's headline metric. **g**, Energy distance against the mean-only limit as the spread λ is restored, closing at $\lambda = 0$, where the energy distance between two point masses is exactly twice the Euclidean distance between their means. **h**, Four per-subpopulation distances 0.20, 0.55, 0.80 and 1.30, the worst-matched state crossed; frames, the weight the aggregate places on each state at $\beta \rightarrow 0$, $\beta = 2$ and $\beta \rightarrow \infty$.

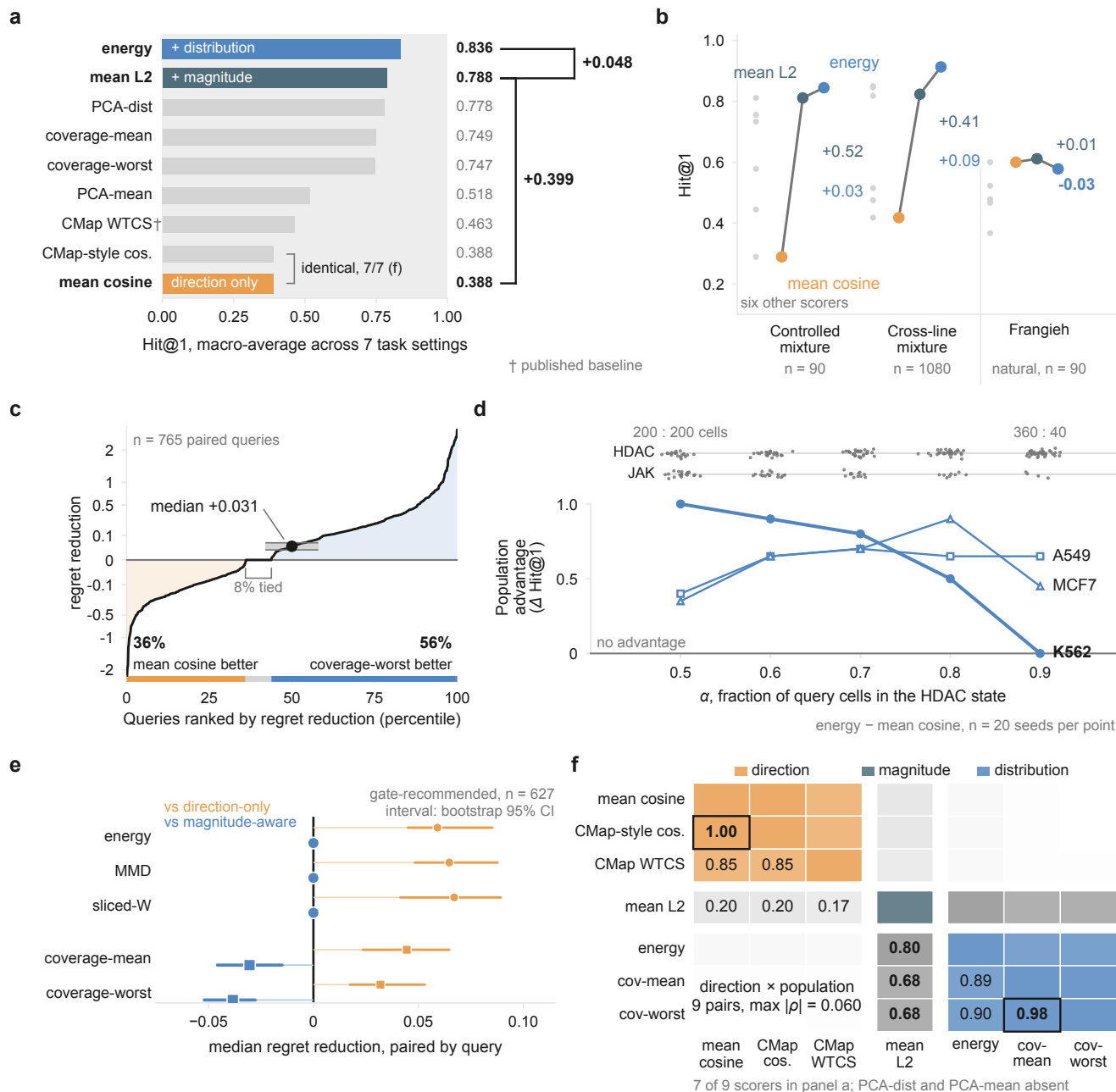


Figure 2: A magnitude-aware mean score recovers most of the gain beyond directional scoring. Points and intervals are defined in Methods. **a**, Hit@1 for nine scorers, macro-averaged over seven task-setting combinations: direction-only mean cosine 0.388, magnitude-aware mean L2 0.788 and energy distance 0.836, so the two steps are +0.399 and +0.048. Dagger, the published CMap connectivity score (WTCS). **b**, The same three scorers per task ($n = 90, 1,080$ and 90 queries); the magnitude step is +0.52 and +0.41 on the two SciPlex3 tasks and +0.01 on Frangieh, where the distributional step is -0.03. **c**, Worst-case subpopulation coverage against direction-only mean cosine over 765 partial-observation queries; curve, the quantile function of the per-query regret reduction, positive favouring coverage (less regret on 56% of queries, more on 36%, ties on 8%). Median +0.031 (band, 95% bootstrap CI +0.017 to +0.049; paired Wilcoxon $P = 1.6 \times 10^{-12}$, mean +0.132), and +0.035 on the 165 queries with a genuinely incomplete library ($P = 6.0 \times 10^{-5}$). **d**, Energy advantage over mean cosine against the minority fraction α ($n = 20$ seeds per point; top, query composition), falling from +1.00 to 0.00 in K562 and not falling in A549 or MCF7 (+0.40 to +0.65; +0.35 to +0.45). **e**, Median regret reduction for five population-level scorers on 627 diagnostic-positive queries. Against direction-only mean cosine all five 95% bootstrap intervals exclude zero (+0.032 to +0.067); against the magnitude-aware mean L2 the three distributional distances have a median of exactly zero and both coverage scores are negative (-0.030, -0.038). **f**, Spearman correlation among seven scorers over 54,180 query-candidate scores from 1,260 queries, an agreement between scoring rules rather than retrieval evidence; shade, $|\rho|$; rings, the closest pair within a tier. The magnitude-aware mean L2 sits with the distributional scores ($\rho = 0.68$ to 0.80), not with the direction-only scores ($\rho = 0.17$ to 0.20).

distances have a median difference of exactly zero and both coverage scores are negative (Fig. 2e).

A magnitude-aware mean score therefore recovers most of the gain beyond directional scoring, and the further increment from retaining the full population is real and much smaller. Both criteria used so far reward correspondence between response populations, so we next ask whether that increment survives biological and functional evaluation.

Population-level gains weaken under biological and functional evaluation

Whether that additional information improves the same candidate rankings depends on how success is defined. We therefore kept the rankings fixed and changed only the criterion used to judge them (Fig. 3).

Mechanism-of-action recovery is the first task-proximal biological criterion. Among the 600 partial-observation queries carrying both response-matching and mechanism annotations, the worst-case subpopulation-coverage scorer improved on direction-only mean cosine by a median of +0.028 under response matching, whereas the identical rankings judged by mechanism-of-action recovery gave a median of 0.000 and a mean of -0.024 (Fig. 3a,b). At the query level, mechanism recovery decreased more often than it increased (40.8% against 35.7%). The same divergence between the two criteria appeared in all five population-level scoring variants, whose response-matching median gains span +0.028 to +0.067 while their mechanism-recovery means span -0.011 to -0.033. Gains under response matching therefore did not consistently carry over to the recovery of biologically related drugs.

A second task-proximal criterion tests how well population-level retrieval preserves minority-state responses. Across 239 task instances, population-level retrieval improved minority-state coverage by only +0.0011 on average, and in 102 of them the two approaches scored exactly the same (Fig. 3c). The small positive effect was concentrated in the constructed cross-cell-line tasks (+0.0037), whereas the remaining settings were near zero or slightly negative. Stratifying by measured response divergence revealed a gradient only on the objective-aligned criterion: the minority-state advantage rose from below zero in the least divergent quartile to about +0.005 in the two most divergent (Fig. 3d), while no divergence stratum reached a mechanism-of-action gain whose interval lay above zero (Fig. 3e). Stronger population differences therefore do not by themselves produce a more biologically useful ranking, and the pre-specified diagnostic gate did not identify the queries that did benefit (Fig. 3f,g; Supplementary Note 1).

Population-level retrieval does carry information that an external functional readout can see. Using GDSC2^{28,29} dose-response profiles from cell lines excluded from the transcriptional retrieval experiment, and removing the shared cell-line sensitivity effect, energy-distance retrieval reached a Spearman correlation of +0.265 with drug-drug functional similarity, against +0.083 for the corresponding mean-signature baseline (Fig. 3h; Supplementary Table 3).

Much of that advantage did not require comparing cellular distributions. Ranking candidates only by how closely their overall response magnitude matched the query reached +0.232 from a single scalar per perturbation, and once that magnitude channel was accounted for the remaining association for energy retrieval was +0.105, positive in all three cell lines (Fig. 3h,i). Scored against absolute GDSC2 potency instead, energy and raw mean cosine turn negative, since matching a weak query's magnitude returns other weak drugs, whereas ranking by candidate magnitude alone turns positive (Fig. 3j). The channel is also directly measurable: across queries the query-candidate energy distance tracks the candidate's own response magnitude at a median Spearman correlation of +0.780 (Fig. 3k). Population-level retrieval therefore carries external functional signal, and most of the signal detected here is recoverable without any cell-to-cell distributional structure.

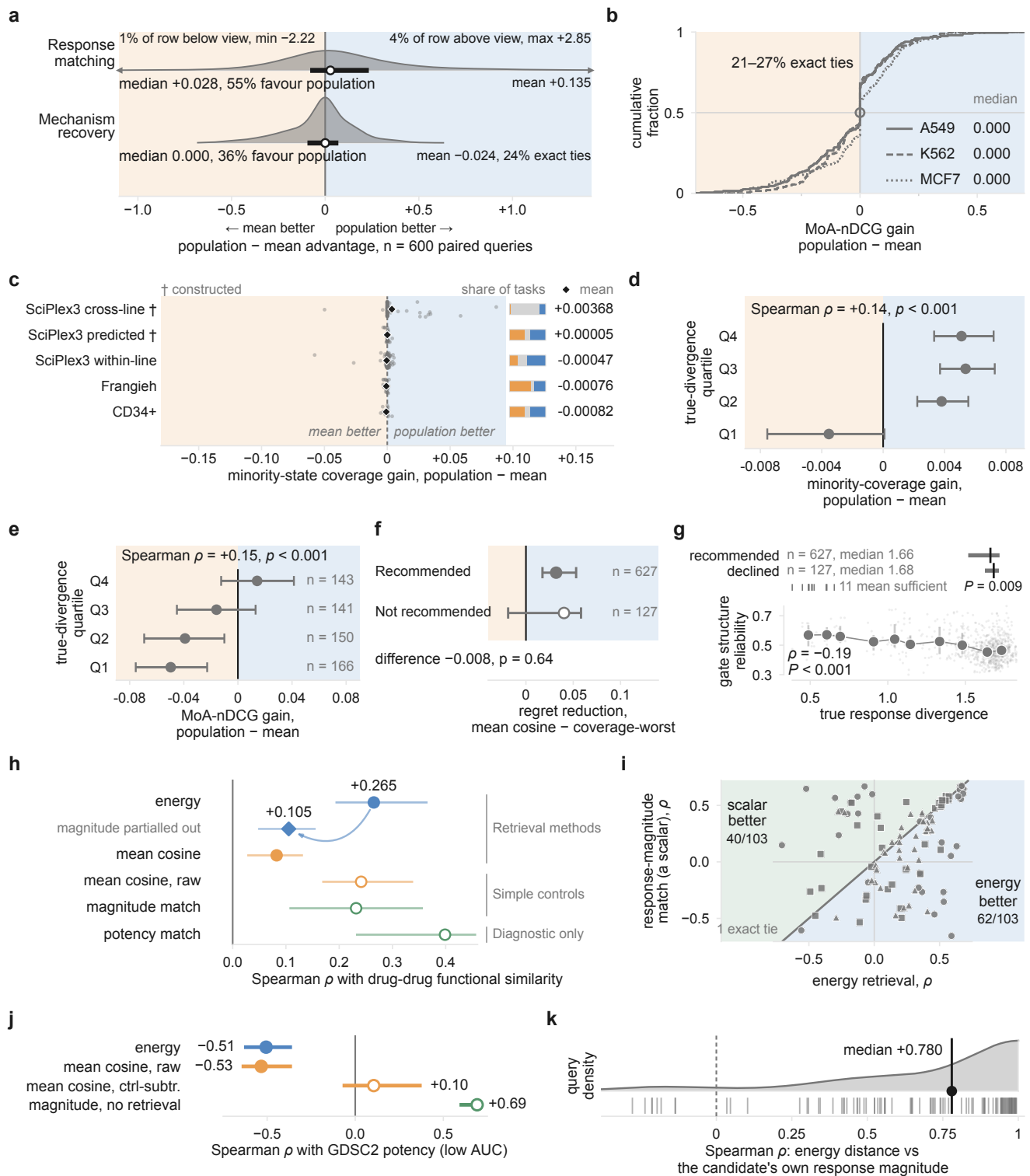


Figure 3: The population advantage weakens, and can change direction, once the judge stops sharing the retrieval objective. Summary points, medians, except in **d** and **e**, where they are quartile means; bars, 95% bootstrap intervals, except in **a**, **g** and **j**, where they are interquartile ranges; cell lines, line style in **b**, marker shape in **i**. **a**, 600 leave-one-drug-out queries judged twice: median +0.028 under response matching against 0.000 under mechanism-of-action recovery (24% ties; Wilcoxon $p = 7.7 \times 10^{-3}$). **b**, Per cell line, medians exactly 0.000 in all three ($n = 200$ each). **c**, Minority-state coverage gain over 239 tasks, 102 of which score identically; dots, individual tasks; diamonds, dataset means; daggers, constructed data. **d**, Population-minus-mean gain by divergence quartile, objective-aligned judge ($n = 765$); quartile means -0.0035 (CI -0.0076 to +0.0001), +0.0038, +0.0054, +0.0051, against medians 0.0000, +0.0004, +0.0012, +0.0020 (Spearman $\rho = +0.14$, $p = 1 \times 10^{-4}$). **e**, The same quartiles under mechanism-of-action recovery ($n = 600$): no interval on the quartile mean lies above zero, the two least divergent wholly below it (-0.050, -0.039; $q = 0.002$, 0.025, Benjamini-Hochberg). With 17 to 32% of queries in each quartile scoring identically under this criterion, the corresponding medians are -0.0062, 0.0000, 0.0000 and 0.0000; the mean is the summary drawn here and the median is the summary quoted in **a** and **b**. **f**, Regret reduction, 627 recommended queries against 127 declined (-0.008, CI -0.032 to +0.056, Mann-Whitney $p = 0.64$). **g**, Below, structure reliability against divergence ($\rho = -0.19$, $P = 1.5 \times 10^{-7}$); above, declined queries are more divergent (1.68 against 1.66, $P = 0.009$). **h**, Ranking against GDSC2 drug-drug similarity ($n = 103$); filled, ranking methods; arc, energy partialled on magnitude. **i**, Per query: energy beats the magnitude scalar on 62 queries and loses on 40 (paired $p = 0.054$). **j**, Rankings against absolute GDSC2 potency: the two rankings computed without control subtraction, energy and raw mean cosine, turn negative (-0.506, -0.533); the control-subtracted mean cosine stays just above zero (+0.105) and candidate magnitude alone is highest (+0.692). **k**, Per-query

170 The apparent value of population-level retrieval therefore depends on how success is measured, and
171 it does so even when the criterion moves beyond transcriptomic response matching.

172 **Evaluator form can reverse the preferred retrieval strategy, and natural heterogeneity leaves little** 173 **reordering headroom**

174 We next asked whether the dependence on the evaluation criterion persists when rankings are judged
175 by a biological measurement that no retrieval method can see.

176 Frangieh Perturb-CITE-seq³⁰ measures RNA and 20 surface proteins in the same 218,331 cells, so an
177 RNA-derived ranking can be judged by a modality withheld from every retrieval method. We built two
178 protein-based evaluators from those same measurements, changing only how they were summarized,
179 and scored the identical two RNA-derived rankings against both (Fig. 4a). The first compared the mean
180 protein responses of each query–candidate pair, and under it mean-signature retrieval outperformed
181 energy-distance retrieval (+0.242 versus +0.222).

182 Against the full distributions of the same protein responses the ordering reversed: energy-distance
183 retrieval reached +0.393 against +0.230 for mean-signature retrieval (Fig. 4b). Energy led under the
184 distribution-based evaluator in all three immune conditions, and the change in which method wins
185 occurred in two of the three. Changing only how the same held-out biological measurements were
186 represented was therefore enough to reverse which retrieval strategy performed better.

187 Part of that reversal needs no distributional comparison. A scalar score that compares only the over-
188 all magnitude of the query and candidate responses rose from +0.125 under the mean-based protein
189 evaluator to +0.276 under the distribution-based one. Energy distance stayed ahead of that control un-
190 der both forms, by +0.098 and +0.117 and in every immune condition separately (Fig. 4c). Response
191 magnitude therefore explains part of the reversal, while energy retains an additional association beyond
192 that scalar channel. The same held-out measurements can favour different retrieval strategies depending
193 on how their response structure is represented.

194 A controlled HIR-Bench construction confirms that subpopulation structure can change the preferred
195 candidate where state-specific preferences conflict, although its analytical boundary did not transfer to
196 the biological datasets (Fig. 4d; Supplementary Note 2).

197 We next asked whether state-dependent perturbation responses are recoverable in naturally hetero-
198 geneous tissue¹⁶. In glioblastoma samples from ZhaoSims2021, the malignant and myeloid compart-
199 ments of each tumour were analysed separately against compartment-matched controls. A supervised
200 probe recovered within-patient drug-response identity at an accuracy of 0.923 against 0.777 for the best
201 unsupervised procedure, a median paired gap of +0.117 against +0.007 in a constructed comparator
202 (Fig. 4e; Supplementary Note 3).

203 Whether that structure is large enough to change which drug should be preferred is a separate ques-
204 tion, and the tumour data constrain it. Across 18 within-patient drug pairs, myeloid-response similarity
205 ranked malignant-response similarity at Spearman $\rho = +0.835$ across the two disjoint compartments,
206 which share no cells (Fig. 4f). The pairs are not independent: 15 of the 18 come from one patient and the
207 remaining three from one patient each, so no P value is quoted and the coefficient describes this cohort
208 rather than estimating a population value. Drug-response similarity was therefore largely preserved
209 across distinct cellular compartments, leaving less room for state-specific structure to reorder candidates
210 than the presence of heterogeneity alone would suggest. This measures preservation of drug ordering
211 rather than retrieval performance, and it is the requirement the intervention-identification task below
212 tests directly: cellular structure must not only exist and be recoverable, but must also change candidate

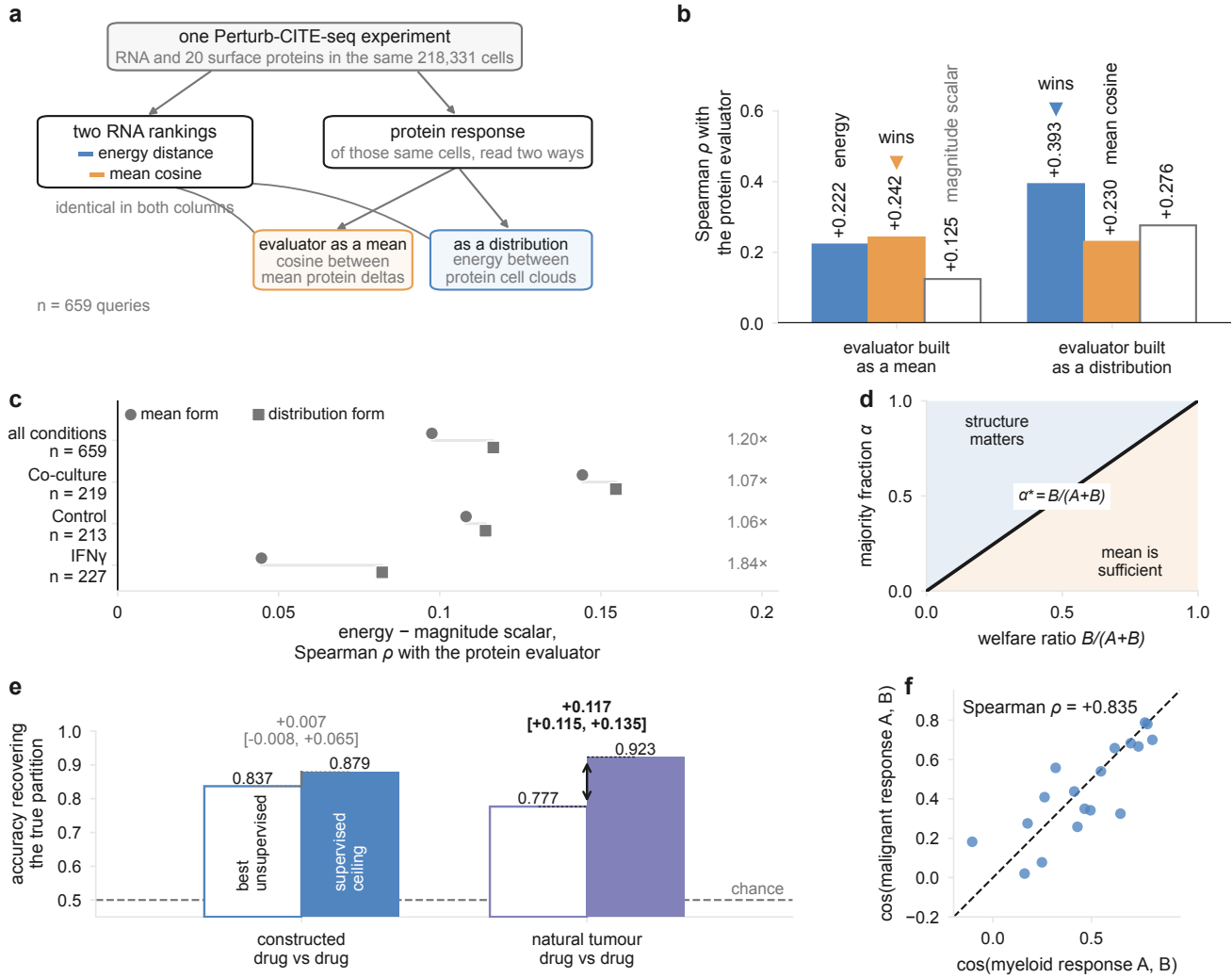


Figure 4: External evaluation is not automatically neutral, and natural heterogeneity does not automatically imply decision reordering. Full definitions and robustness analyses are in Supplementary Notes 2 and 3. **a–c**, One external modality judging the same rankings two ways. **a**, One Perturb-CITE-seq experiment measures RNA and 20 surface proteins in the same 218,331 cells; the two RNA rankings and the control-centred protein responses are held identical while only the statistical form of the protein evaluator changes. **b**, Spearman ρ of three RNA-side scores against that evaluator, built once as a cosine between mean protein deltas and once as an energy distance between protein cell clouds; each bar is the median over $n = 659$ queries of a per-query rank correlation across that query's candidate knockouts. Open bar, the response-magnitude control, which ranks candidates but compares no distributions. **c**, How far energy exceeds the magnitude control under each form, pooled and within each immune condition (ratios 1.06 to 1.84); no uncertainty is shown, because the source records one correlation per condition per form. **d**, The HIR-Bench boundary $\alpha^* = B/(A+B)$ separates settings in which the mean is sufficient from settings in which subpopulation structure can change the preferred candidate, without identifying them in biological data (Supplementary Note 2). **e, f**, The same question in four glioblastoma tumours, each split by cell identity into disjoint malignant and myeloid compartments that share no cells and see the same drug and matched control. **e**, Response-identity recoverability of within-patient drug-response structure; open bars, best unsupervised method; filled bars, supervised probe; median paired gap +0.117 (95% patient-level cluster bootstrap +0.115 to +0.135, $n = 36$ splits), against +0.007 in a constructed comparator. **f**, Myeloid-response similarity against malignant-response similarity across those disjoint compartments, on which no retrieval was performed (Spearman $\rho = +0.835$, $n = 18$ drug pairs, 15 of them from a single patient; dashed line, equal similarity).

213 preference.

214 Population information changes an intervention decision when candidate responses are observed

215 A retrieval system that is to propose interventions must rank compounds whose effect in the query con-
216 text has not been measured. We therefore ran the same comparison as a source-to-target intervention-
217 identification task on Tahoe-100M¹⁵: a held-out cell line’s untreated cells and a desired treated popula-
218 tion enter, and a ranking of a fixed 92-compound library leaves, with the true intervention’s rank as the
219 outcome (3,992 queries over 44 cell lines; Fig. 5a). Because the target is an observed response produced
220 by one library compound, the task measures intervention identification rather than therapeutic utility.
221 It was run twice, once with candidate responses observed within the query line and once with them
222 predicted from the other 43 lines, which separates the value of population information from the cost of
223 forecasting it.

224 With candidate responses observed, the three scoring rules of Fig. 1a again form a ladder. Mean re-
225 ciprocal rank was 0.945 for direction-only mean cosine, 0.965 once response magnitude was restored and
226 0.976 for the energy distance between populations, so the step to the magnitude-aware mean is +0.0201
227 and the step from it to the population +0.0102 (Fig. 5b). That last step is not a property of population
228 scoring as such. Of the two secondary population scores specified for this task, maximum mean discrep-
229 ancy reproduces the energy result (MRR 0.975, +0.0101 over the magnitude-aware mean), whereas sliced
230 Wasserstein distance reaches 0.922 and sits below both mean baselines (−0.0230 against direction-only
231 matching, 95% CI −0.0301 to −0.0166; −0.0431 against the magnitude-aware mean, −0.0509 to −0.0359),
232 so two of the three population scores clear both mean baselines here and one does not. The advantage
233 reaches individual decisions: against the magnitude-aware mean, population scoring put the true drug
234 first on 2.0% of queries where the mean route did not and broke a correct choice on 0.5%, a fourfold
235 asymmetry, and against direction-only matching the two rates are 5.6% and 1.1% (Fig. 5c).

236 The pooled oracle gain is limited by the headroom that mean matching leaves. On 3,110 of the
237 3,992 queries the direction-only mean route already ranks the true drug first at every seed, where the
238 population route can only lose and does, by −0.0036 on average; the remaining 882 queries more than
239 account for the pooled +0.0303, gaining +0.1501 (Fig. 5d). Headroom of that kind correlates with the
240 population route’s gain at Spearman $\rho = +0.79$, but a permutation null that preserves the ranking
241 ceiling reproduces it at +0.786 (+0.767 to +0.805; Fig. 5e, Methods), so the correlation measures the
242 room available for improvement rather than which queries benefit.

243 We next asked which measurable query properties explain the remaining advantage. A query drug’s
244 own treated cells are separable from vehicle cells of the same line at a median accuracy of 0.800 when the
245 labels are given and 0.555 without them, on a balanced two-class problem where 0.5 is chance (Fig. 5f),
246 and drug-by-state interaction was measured by a cross-fitted statistic that is zero in expectation under
247 an additive null and largely free of the cell-count and magnitude dependence a naive response cosine
248 carries (Fig. 5g; Methods). Conditional on the mean route’s own reciprocal rank, response detectability
249 is the only one of the three whose interval excludes zero (+0.0091, 95% CI +0.0060 to +0.0121, $n = 3,618$
250 queries in 44 cell-line clusters); neither the interaction share (+0.0015, $p = 0.21$) nor its reproducibility
251 across disjoint halves (−0.0011, $p = 0.21$) is associated with the population route. Substituting the
252 naive cosine for the cross-fitted statistic reverses the sign of the association (−0.0096, 95% CI −0.0167 to
253 −0.0024; Fig. 5h). Population-level information therefore has measurable value at the oracle, and little
254 room in which to act.

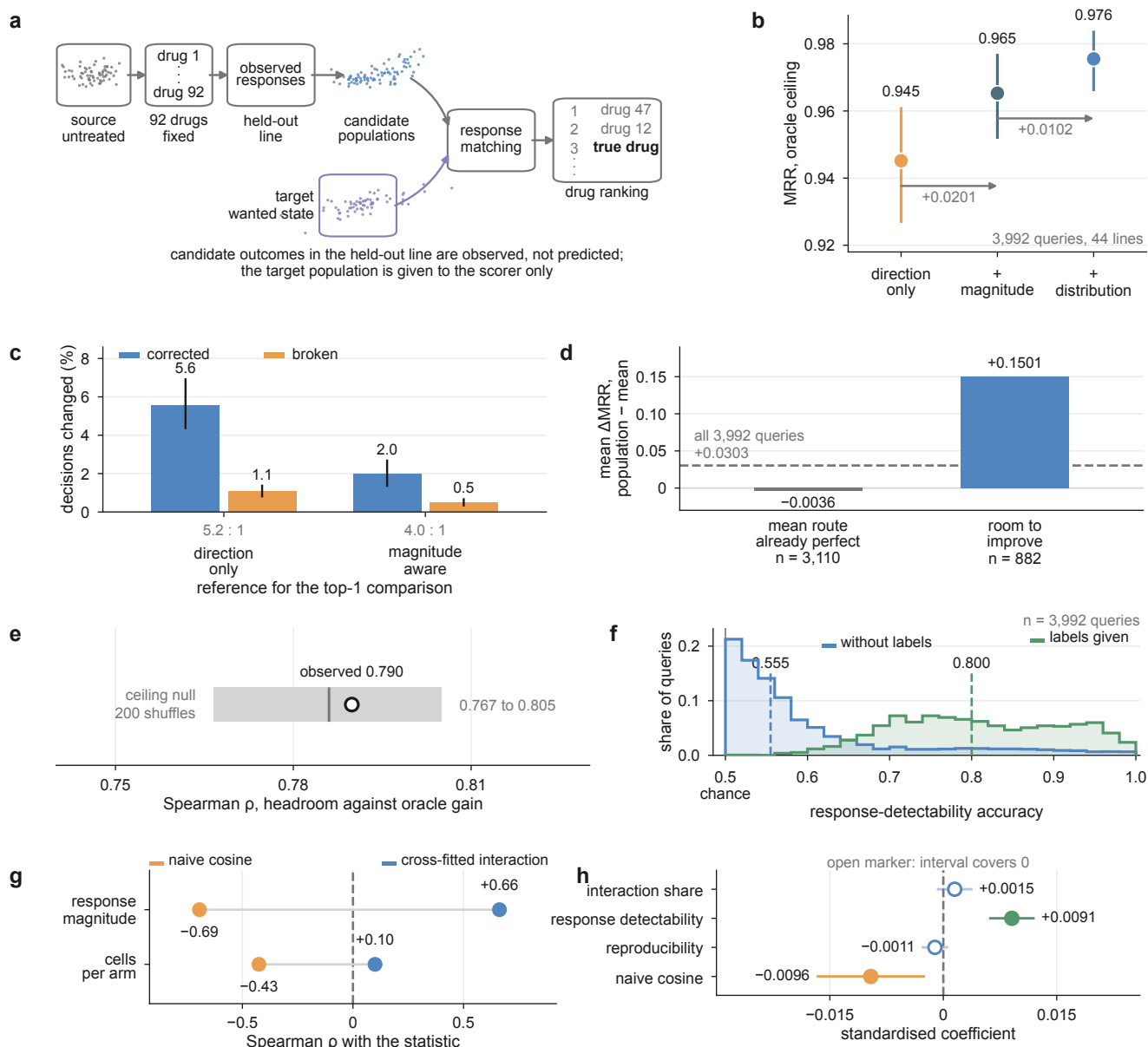


Figure 5: Population structure adds a small oracle retrieval gain beyond response magnitude, and most of it comes from a fifth of the queries. Source-to-target intervention identification on Tahoe-100M: 3,992 queries over 44 cell lines and a fixed 92-compound library, with candidate responses *observed* within the query line throughout and predicted from the other 43 lines in Fig. 6; **a** is schematic and energy distances use the unbiased U-statistic estimator. **a**, A held-out line's untreated cells and a desired treated population enter and a ranking of all 92 compounds leaves; the target population is the query, not a candidate response. **b**, Mean reciprocal rank for the three scoring rules of Fig. 1a, with 95% bootstrap intervals: 0.945, 0.965 and 0.976, so the two steps are +0.0201 and +0.0102. **c**, Top-1 decisions changed against each mean reference, ratios above the pairs; corrected, the population route puts the true drug first where the reference did not, and broken is the reverse. **d**, The same +0.0303 pooled advantage split by whether the mean route left any room; on 3,110 of 3,992 queries it already ranks the true drug first at every seed. **e**, Spearman correlation between that headroom and the population gain, against a null that reshuffles the population route's reciprocal ranks within each context, keeping both marginals, the $RR \leq 1$ ceiling and the clustering. **f**, Single-cell response detectability, separating a query drug's own treated cells from vehicle cells of the same line at up to 300 cells per arm; median accuracy 0.800 with the labels given and 0.555 without them, on a balanced two-class problem where 0.5 is chance. **g**, Each interaction statistic against the two quantities it must not measure: the naive $1 - \cos(r_1, r_2)$ runs at -0.69 with response magnitude and -0.43 with cells per arm, the cross-fitted S_{int} at +0.66 and +0.10. **h**, Standardised coefficients from regressions of the population route's own reciprocal rank on each explainer, with the mean route's reciprocal rank as a covariate; the first three rows are one fit ($n = 3,618$ queries in 44 cell-line clusters) and the naive cosine a separate one ($n = 3,620$). Filled markers, coefficients whose interval excludes zero.

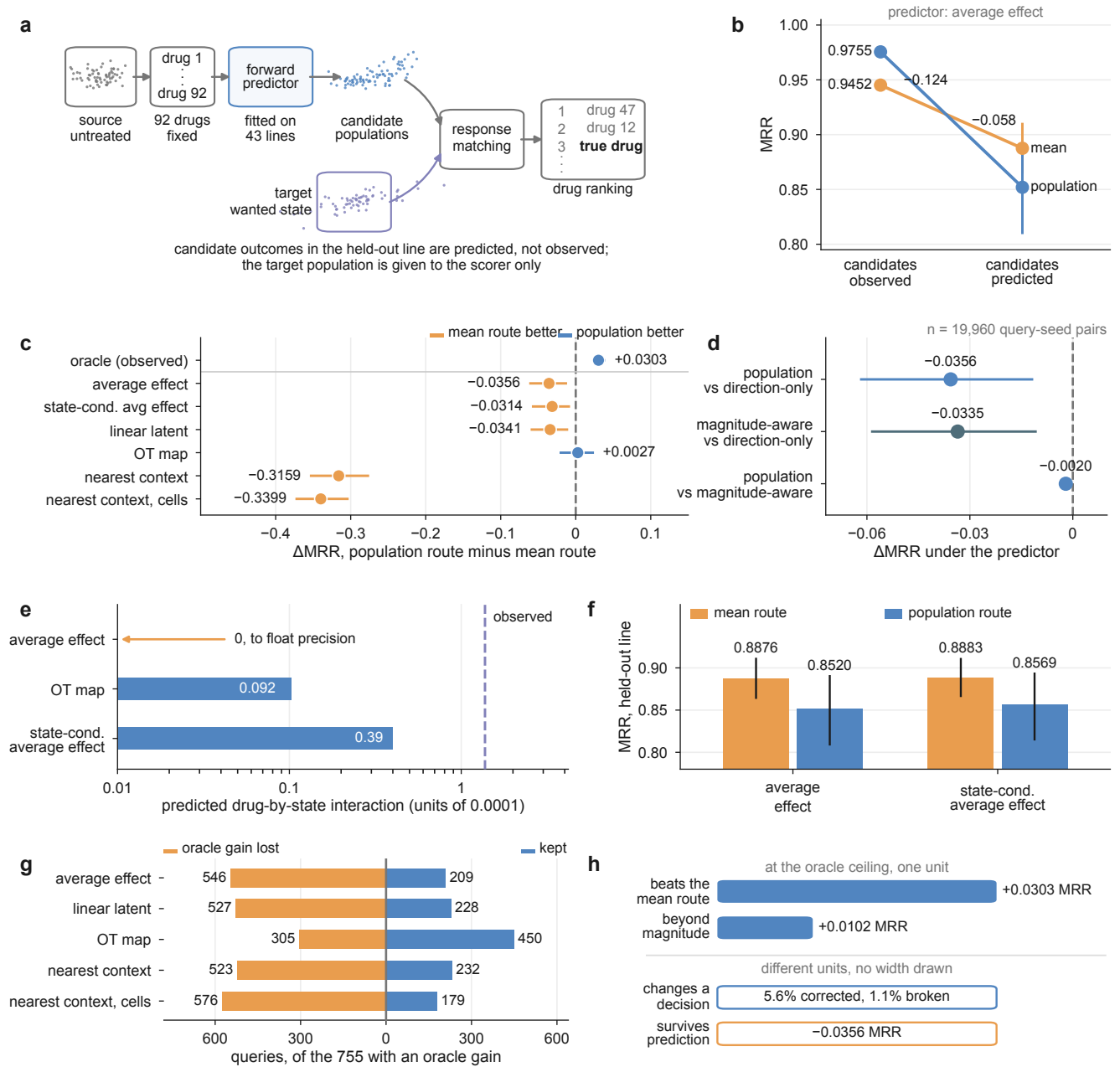


Figure 6: The oracle advantage does not survive forward prediction, and the deficit is a magnitude error rather than a distributional one. The same task, queries, library and split as Fig. 5, with candidate responses in the held-out line produced by a forward predictor fitted on the other 43 lines; panel **a** is schematic and intervals are 95% bootstrap intervals over cell-line clusters. **a**, The task with that one station replaced; every other element is drawn as in Fig. 5a. **b**, Both routes travelling from observed to predicted candidates under the primary average-effect predictor; the mean route gives up 0.058 MRR and the population route 0.124, and the two cross. **c**, Δ MRR of the population route over direction-only mean matching, at the oracle and for six predictors; under an additive predictor the oracle gain inverts rather than attenuating. **d**, The same predictions judged three ways: the population route's -0.0356 deficit against direction-only is matched by the magnitude-aware mean's -0.0335 , and -0.0020 remains once magnitude is controlled. **e**, Median drug-by-state interaction in each predictor's own output against the cross-fitted value measured in real cells (dashed); an additive predictor's is zero to float precision, an algebraic property of the model class. **f**, A state-conditioned average-effect predictor is the first to generate real interaction without giving up mean-route accuracy (panel **e**), and it does not close the gap. **g**, The same attrition counted over the 755 queries on which the oracle population route had any gain: 209 keep a gain under the primary average-effect predictor and 546 do not. Gains also appear where the oracle had none, 343 for that predictor and 1,165 for the Gaussian OT map, on a different denominator and, for that map, on a mean route reaching only 0.665. **h**, The budget across both figures; the first two rows share the oracle MRR unit and are drawn to scale, and the last two are a rate and a signed gain and carry no width.

255 The oracle advantage does not survive forward prediction

256 Replacing the observed candidate responses with predicted ones removes the advantage and reverses its
257 sign (Fig. 6a). The primary average-effect predictor costs the mean route 0.058 MRR and the population
258 route 0.124, and the two cross (Fig. 6b). Under it the population route sits -0.0356 below direction-
259 only mean matching (95% CI -0.0620 to -0.0115), so the oracle gain does not attenuate but changes
260 sign (Fig. 6c). None of the six forward predictors evaluated delivers a population advantage, and the
261 Gaussian optimal-transport map is the only one whose point estimate is not negative ($+0.0027$, interval
262 covering zero).

263 The dominant deficit is a magnitude error rather than a distribution-specific one. On the same pre-
264 dictions the magnitude-aware mean loses -0.0335 to direction-only matching, almost the whole of the
265 population route’s deficit, although it reads no distribution at all; against the magnitude-aware mean
266 itself the population route sits only -0.0020 below (-0.0036 to -0.0005 ; Fig. 6d). An additive predictor
267 translates the query line’s own vehicle cells by an effect averaged over the other 43 lines, so it reproduces
268 that population’s heterogeneity exactly while generating no drug-by-state interaction at all, an algebraic
269 property of the model class rather than a training failure (Fig. 6e; Supplementary Note 4).

270 A state-conditioned average-effect predictor is the first to generate real interaction while preserving
271 the mean route’s MRR (0.8883 against 0.8876), and it still leaves the population route behind (-0.0314 ,
272 -0.0584 to -0.0076 ; Fig. 6f). Generating state-specific interaction is therefore necessary for the popula-
273 tion route but not sufficient. The same attrition is visible query by query: of the 755 queries on which
274 the oracle population route had any gain, the average-effect predictor keeps one on 209 and loses it on
275 546 (Fig. 6g).

276 Together the two arms locate where population-level information enters and where it is lost (Fig. 6h).
277 With responses observed it is real, it reaches a ranking decision, and it is bounded by how good mean
278 signatures already are. With responses predicted it does not survive: the predictors that best preserved
279 mean-level retrieval were additive and generated no state-specific perturbation response, and introduc-
280 ing state conditioning restored that interaction without restoring a population-level advantage.

281 Discussion

282 The results trace one chain of attrition rather than a checklist. Response populations carry retrievable
283 information; a smaller part of it exceeds response magnitude; a smaller part again changes which can-
284 didate is preferred; and almost none of that survives once candidate responses must be predicted rather
285 than observed. The binding constraint changes along that chain: at the oracle it is the ranking headroom
286 that mean matching leaves, because mean matching already ranks the true compound first on most
287 queries, and once candidate outcomes are unobserved it is the fidelity of the forward predictor.

288 Single-cell perturbation atlases^{13–15} change the representation available for drug retrieval: a pertur-
289 bation can now be described by a population of cellular responses rather than by a single averaged
290 signature. Across the datasets and evaluators examined here, retaining the full population increased
291 the information available to the retriever, but only a smaller component beyond response magnitude
292 reached the ranking, and it did so under response matching, where Fig. 2e shows that it adds nothing to
293 the magnitude-aware mean under the regret criterion; under mechanism-of-action recovery the whole
294 population advantage does not survive (Fig. 3a,b,e). The central distinction is therefore not mean versus
295 distributional retrieval, but information that is available versus information that changes the decision.
296 Classical perturbation signatures and single-cell populations are points on a continuum of retained re-
297 sponse structure, and the relevant question is how much of that structure the decision at hand requires.

298 This distinction has direct implications for artificial intelligence virtual cells^{1,2}. A virtual cell that
299 accurately predicts an average perturbation response can still fail to provide the information needed for
300 population-level intervention selection. A model that applies the same perturbation effect to every cell
301 can preserve heterogeneity in the starting population yet, by construction, cannot generate state-specific
302 differential responses; the limit is exact for strictly additive predictors, although more flexible latent
303 models are not algebraically constrained to it (Supplementary Note 4). A downstream population-aware
304 retriever cannot recover variation that the predictor never generated. Reproducing state-dependent dif-
305 ferential response is therefore necessary to exploit that particular source of population information, and
306 it is not sufficient: introducing state conditioning restored measurable drug-by-state interaction while
307 preserving mean-level performance, yet did not restore a population-level retrieval advantage (Fig. 6e,f).
308 The same experiment identifies where to look next, because the magnitude-aware mean loses almost ex-
309 actly what the population route loses, so the dominant remaining error is predicted response magnitude
310 rather than population shape (Fig. 6d). That is a concrete target for virtual-cell modelling.

311 Two limits bound these conclusions. The response and interaction diagnostics examined here did
312 not yield a reliable query-level rule for when population scoring helps, and state-dependent differential
313 response is only one of the forms population structure could take, so a gain arising from another form is
314 not excluded. These results also suggest a practical principle for evaluating population-aware retrieval.
315 The statistical representation of an external readout can itself change which method ranks higher^{27,31},
316 so validation should ask whether the added cellular information changes an outcome relevant to the
317 intended use, and should compare against simpler channels such as response magnitude²³ before at-
318 tributing a gain to population structure itself. The functional readouts analysed here remain indirect
319 and do not establish therapeutic benefit. More broadly, a richer cellular measurement justifies a richer
320 retrieval method only when the information it adds survives to the decision that method exists to make.

321 **Methods**

322 **Study design and data resources**

323 This study asks when retaining the full population of single-cell perturbation responses improves re-
324 trieval beyond a conventional mean expression signature. The analyses were therefore organized around
325 four distinct roles: observed-population retrieval, evaluation against biological and functional criteria,
326 measurement of naturally occurring response heterogeneity, and large-scale assessment of the condi-
327 tions under which population-level information can be useful. These roles were analysed separately
328 rather than pooled into a single benchmark.

329 The primary observed-population retrieval analyses used SciPlex3¹⁴, comprising 276,325 cells across
330 three cancer cell lines (A549, K562 and MCF7), 188 compounds shared across the three contexts and
331 a 2,000-gene expression space. SciPlex3 supports both within-context retrieval and controlled cross-
332 context constructions because the same compound library is measured across all three cell lines. Frang-
333 ieh melanoma Perturb-CITE-seq³⁰ spans Control, IFN γ and Co-culture conditions with 239 CRISPR
334 knockouts. Both analyses read the same RNA matrix under two filters: the retrieval analysis uses the
335 218,023 cells left after dropping nine knockout targets with fewer than 50 cells, and the evaluator-shape
336 experiment uses all 218,331 cells of the RNA-protein barcode intersection, of which 215,915 contribute
337 to at least one condition mean once the 60-cell floor per knockout is applied. We use this dataset as
338 an independent perturbation-retrieval setting and, because matched surface-protein measurements are
339 available, for the evaluator-shape experiment described below, in which that protein modality is held
340 out from every retrieval method and used only to judge. It is a genetic-perturbation test of the same

retrieval formulation rather than a drug-retrieval benchmark. CD34+ haematopoietic progenitor data (GSE306429)³² contain 33,984 cells and 36 compounds and are used as a negative-control setting for population-level retrieval.

Three additional resources serve roles outside the primary retrieval benchmark. ZhaoSims2021¹⁶ contains drug-treated patient-derived glioblastoma tissue and is used to test whether state-dependent responses and their recoverability are present in naturally heterogeneous tissue; no retrieval performance is reported from this cohort. Tahoe-100M¹⁵ serves two distinct roles: it provides larger-scale measurements of the same information requirements under intrinsic within-cell-line heterogeneity, and it is the substrate for the source-to-target intervention-identification benchmark of Figs. 5 and 6. We analyse plate 3 only, which comprises 4,158,278 cells across 50 cell lines and 93 compounds, using the released 2,304-gene log1p-transformed representation without refitting the preprocessing pipeline. Finally, GDSC2 dose-response profiles provide an external functional readout for drug-drug similarity and are joined to the SciPlex3 compound panel independently of the transcriptomic retrieval scores.

Observed responses, predicted responses, external functional measurements and natural-tissue analyses therefore answer different questions in this study. Results are not combined across these resources as though they represented independent replicates of one benchmark. Dataset scale, inclusion criteria, experimental role and query construction are summarized in Supplementary Table 2.

Preprocessing and construction of perturbation-response populations

Single-cell expression analyses were performed in log-transformed expression space using the processed gene sets defined for each dataset. SciPlex3, Frangieh and CD34+ were analysed in 2,000-gene representations. For SciPlex3 and Frangieh these genes were selected in this study rather than inherited from the released objects: counts were normalized to a total of 10^4 per cell and log1p-transformed, 2,000 highly variable genes were chosen by the Seurat criterion³³ on a subsample of cells, and the final matrices were then rebuilt from raw counts on those genes using all-gene size factors so that normalization is applied once. SciPlex3 was additionally downsampled to at most 120 cells per compound, cell line and dose, and 3,000 control cells per cell line. The CD34+ tensor was received in processed form and the feature-selection procedure applied to it is not documented by the source. For ZhaoSims2021, counts were normalized to a total of 10^4 per cell, log1p-transformed and reduced to 2,000 highly variable genes. Tahoe-100M was used in the released preprocessed form (2,304 highly variable genes, log1p transformed), with no additional feature selection or representation fitting. Dataset-specific filtering and minimum-cell requirements for individual analyses are given in the corresponding Methods sections below.

For a perturbation d measured in cellular context c , let

$$X_{d,c} = \{x_1^{(d,c)}, \dots, x_{n_{d,c}}^{(d,c)}\}$$

denote its treated single-cell expression profiles and let

$$\mu_{0,c} = \frac{1}{n_{0,c}} \sum_{j=1}^{n_{0,c}} x_j^{(0,c)}$$

denote the mean of the matched control population. The conventional perturbation signature is the

376 control-referenced mean response

$$\bar{d}_{d,c} = \frac{1}{n_{d,c}} \sum_{i=1}^{n_{d,c}} x_i^{(d,c)} - \mu_{0,c}.$$

377 Mean-signature retrieval operates only on $\bar{d}_{d,c}$.

378 Population-level retrieval retains the corresponding control-referenced cellular responses,

$$P_{d,c} = \left\{ x_i^{(d,c)} - \mu_{0,c} \right\}_{i=1}^{n_{d,c}},$$

379 so that the mean of $P_{d,c}$ is exactly the conventional mean perturbation signature while variation among
 380 its cells remains available to the retrieval score. Both arms are therefore built from the same treated
 381 cells, the same matched control, and the same query and candidate libraries. The energy distance and
 382 the mean L_2 depend only on distances between cells and are unchanged by subtracting a control that a
 383 query and its candidates share, so for those two scores the referencing is a change of origin rather than
 384 of comparison; the mean cosine is not invariant in this way and is computed on the control-referenced
 385 means. Where a query and its candidates instead carry different matched controls, as the two single-
 386 mode candidates of the two-context mixture task do, that difference enters the mean scores and not the
 387 population distance.

388 Controls were matched to the biological context in which the perturbation was measured. For anal-
 389 yses that explicitly compare responses between cellular states or compartments, each state is instead
 390 referenced to its own matched control population so that pre-existing differences between states can-
 391 not be mistaken for differential perturbation responses. This state-specific construction is used for the
 392 differential-response analyses in predicted populations, patient-derived glioblastoma and Tahoe-100M
 393 and is defined in the corresponding sections below.

394 Cell sampling was performed only after the response populations had been defined. Minimum cell
 395 numbers, per-condition caps and random seeds differ by experiment and are reported with the corre-
 396 sponding analysis rather than imposed globally. Protein measurements used for the evaluator-shape
 397 experiment were processed separately, including barcode-based RNA-protein matching and protein-
 398 specific normalization, as described below.

399 **Population-to-population retrieval and signature baselines**

400 For a query response population Q and a library of candidate response populations $\{P_d\}_{d=1}^D$, retrieval
 401 assigns each candidate a similarity or distance to the query and orders the candidate library accord-
 402 ingly. Population-level methods operate directly on the empirical response populations defined above,
 403 whereas signature-based methods first reduce each population to its control-referenced mean response.
 404 Query construction and candidate libraries are kept identical when comparing the two representations.

405 **Population-level retrieval.** We evaluated energy distance³⁴ (the primary population score), max-
 406 imum mean discrepancy³⁵, sliced Wasserstein distance³⁶, and the mean and worst-case limits of a
 407 subpopulation-coverage score. All rank candidates from smallest to largest distance. For a candidate
 408 population $P = \{x_1, \dots, x_m\}$ and a query population $Q = \{y_1, \dots, y_n\}$ the energy distance is estimated
 409 as

$$\hat{\mathcal{E}}(P, Q) = \frac{2}{mn} \sum_{i,j} \|x_i - y_j\| - \frac{1}{m(m-1)} \sum_{i \neq i'} \|x_i - x_{i'}\| - \frac{1}{n(n-1)} \sum_{j \neq j'} \|y_j - y_{j'}\|,$$

the Euclidean norm itself rather than its square. Excluding the self-pairs makes this the unbiased U-statistic form; the biased V-statistic divides the two within-population terms by m^2 and n^2 instead, which adds each population's own mean within-population distance divided by its cell count and therefore shifts candidates of different size or dispersion by different amounts. Every energy distance reported here is the U form except the PCA-latent population score of Fig. 2a, which is computed by a separate routine in the V form; the two forms are compared arm by arm in the code release. The population distances are computed on normalized expression rather than on control-referenced responses, whereas the mean-signature scores subtract the matched control mean. Fixed cell caps, penalties for undersized populations and the implementation settings for secondary scores are reported in Supplementary Methods.

Subpopulation-coverage score. The query is partitioned into K components ($k = 2$ throughout) and each candidate cell is assigned to the nearest query component. In the observed-population experiments the partition is the construction label of the two response modes and assignment is by Euclidean nearest centroid; in the partial-observation experiments, where no such label exists, the partition is k -means on the query cells with three initializations and a fixed seed. Writing d_k for the base distance between the query cells and the candidate cells of component k , the coverage distance and its ranking score are

$$D_\beta = \frac{1}{\beta} \left(\log \sum_{k=1}^K e^{\beta d_k} - \log K \right), \quad S_\beta = -D_\beta.$$

The base distance d_k is one of energy, MMD or sliced Wasserstein, so coverage is a family built on the same kernels rather than an independent metric. A candidate that populates no cell of a query component receives the penalty distance for that component, which is how the score charges a candidate for leaving part of the query uncovered.

As $\beta \rightarrow 0$, D_β averages subpopulation distances; as $\beta \rightarrow \infty$, it approaches the worst-matched component: 0.7125 and 1.300 for the four per-subpopulation distances drawn in Fig. 1h. For $K = 1$ it equals the global distance. Coverage-mean and coverage-worst are these two limits; numerical implementation details are given in Supplementary Methods.

Mean-signature baselines. For each response population $P_{d,c}$, the corresponding mean signature is $\bar{d}_{d,c} = \text{mean}(P_{d,c})$. Mean-signature retrieval ranks candidates using either cosine similarity or Euclidean distance between these vectors. The implemented CMap-style cosine baseline applies cosine similarity to the same control-referenced mean differential-expression vectors and is therefore numerically identical to mean-cosine retrieval (Spearman $\rho = 1.000$ over 54,180 query-candidate scores). The canonical L1000 weighted connectivity score (WTCS)¹² is included as a separate signature-based baseline. WTCS uses rank-based enrichment of the mean-collapsed expression response and is related to, but not algebraically identical to, the cosine implementation used here; it gives a different Hit@1 (0.4635).

To test whether the comparison depended on the original gene space, we also evaluated a PCA representation. One basis was fitted per retrieval query, on the query cells pooled with the cells of all 43 candidate populations of that query, the held-out correct candidate included, uniformly subsampled without replacement to 4,000 cells at a fixed seed and reduced to 30 components. The fit is therefore transductive but unsupervised: no candidate identity, ground-truth label or subpopulation label enters it, the correct candidate contributes on the same terms as the other 42, and the query cells are disjoint

from the correct candidate’s cells by construction. Control cells do not enter the fit, each control having already been collapsed to a mean vector that is projected through the fitted basis. The same comparison was then performed either between latent-space means or between the corresponding latent response populations.

Connection between mean and population retrieval. To formalize the relationship between the two representations, we write a response population as

$$X = \mu + \lambda\epsilon, \quad E[\epsilon] = 0,$$

where μ is the mean response, ϵ represents within-population variation and λ controls its magnitude. As $\lambda \rightarrow 0$, the response population collapses to a point mass at μ , and population comparison therefore loses all information beyond the mean. The mean *representation* is the endpoint of this variance-sensitivity construction. The score obtained at that endpoint is a distance between two mean vectors: for the energy distance between two point masses it is exactly $2\|\mu_Q - \mu_d\|_2$, which is the magnitude-aware mean score and not the cosine. Which score reads the mean is therefore a separate choice from how much of the population is retained, and the cosine is a further coarsening of the same representation that discards $\|\mu_Q - \mu_d\|_2$ altogether. Figure 2a measures what that coarsening costs.

The continuum refers to the amount of within-population response structure available to the scorer, not to an equivalence between otherwise different scoring functions: WTCS remains a distinct rank-based signature score, and energy, MMD and sliced Wasserstein retain their own distance definitions.

Response-matching and partial-observation evaluation

Observed-population retrieval. When complete candidate response populations were available, retrieval was evaluated over heterogeneous queries. Each query is a population built by combining two response modes at mixing weight $\alpha \in \{0.5, 0.7, 0.9\}$, and the correct answer is the candidate that reproduces both modes at the same weight, built from the held-out half of the same cells so that no cell is shared between a query and its answer. Two further candidates carry one mode each, and 40 distractor populations complete a 43-candidate library.

The three tasks differ in what the two modes are. In the controlled SciPlex3 task they are the HDAC- and JAK-associated response sources within a single cell line at $10 \mu\text{M}$, each pooling every compound annotated to that class, so the two single-mode candidates are HDAC-only and JAK-only populations and each distractor is an individual compound of another class. In the cross-line and Frangieh tasks they are one perturbation observed in two biological contexts, two cell lines or Control and $\text{IFN}\gamma$, so the two single-mode candidates are that perturbation observed in one context only and each distractor is a different perturbation mixed across the same two contexts at the same weight; cross-line perturbations pool all measured doses. The two single-mode candidates are each referenced to their own single-context control, whereas the correct answer and every distractor are referenced to the α -mixed control.

Query perturbations in these two tasks are not arbitrary. Among perturbations that pass a split-half reliability floor in both contexts and a cross-context response cosine below 0.9, the analysis uses the most divergent: 12 drugs for each cell-line pair and 6 Frangieh knockouts. These two tasks therefore measure retrieval where cross-context divergence is largest rather than on a representative perturbation panel.

Populations contain 400 cells in the controlled task and 200 in the other two, sampled without replacement where the source pool allows it and with replacement otherwise. Within one task setting every query is drawn from a single fixed split of that setting’s cells and differs only in the perturbation

or mechanism sampled, in α and in the resampling seed, so queries within a setting are not independent replicates. The controlled and cross-line tasks use ten resampling seeds and the Frangieh task five.

Hit@1 is reported as an unweighted macro-mean over the seven task–setting combinations, so a 30-query combination counts as much as a 360-query one. Weighting instead by query count gives 0.884 for energy distance and 0.421 for mean cosine, and PCA-space retrieval reaches 0.778 against 0.518 for the corresponding mean-based score.

Hit@1 is the fraction of queries in which no candidate scores strictly higher than the both-mode candidate; its rank is one plus the number of strictly higher-scoring candidates, and the same definition underlies Hit@5, MRR and nDCG@10. That convention would resolve an exact tie in the both-mode candidate’s favour, but no tie arose: across the 1,260 queries and all nine scorers the both-mode candidate is never tied with another candidate at any rank, so every Hit@1 reported here is identical under optimistic, pessimistic and fractional tie handling. The partial-observation protocol below is scored differently: it ranks by a label-blind stable sort and its Hit@1 is the best rank attained by any member of the query’s mechanism or target set, not the rank of a single correct candidate. Ranking a single-mode candidate first counts as a miss, including in the cross-line and Frangieh tasks, where that candidate carries the correct perturbation observed in one of the two contexts. The criterion therefore scores the composition of the retrieved population as well as the mechanism or perturbation it was drawn from. Comparisons between population-level and mean-signature retrieval use the same queries and candidate libraries. Aggregate Hit@1 values are reported as unweighted macro-means over the seven task–setting combinations used in the observed-population analysis; query-weighted averages are reported as a sensitivity analysis so that the largest task does not determine the primary summary.

Partial-observation protocol. We additionally evaluated retrieval when only part of each candidate response population was available to the retriever. This setting was designed to separate the information used to construct a ranking from the criterion subsequently used to evaluate that ranking. For each query, the population-level and mean-signature methods rank the same candidate library once; the resulting rankings are then retained unchanged while different evaluation criteria are applied.

Three splits construct the partial observation, all on SciPlex3, and in each the hidden material never enters the candidate library³⁷. In *leave-drug-out*, one drug is hidden, the query is that drug’s own treated population (up to 200 cells) and the library is every other drug in the context. In *leave-MoA-out*, an entire annotated mechanism class is removed from the library and the query is one drug drawn from the removed class, so recovery requires a same-target surrogate among the remaining drugs. In *partial-library*, a fraction of the drugs is treated as measured, the query is built from the pooled responses of the unmeasured drugs, and the retriever selects the next batch of candidates. In all three, query states are obtained by k -means with $k = 2$ on the query cells and the minority state is the smaller cluster; candidate populations are capped at 120–150 cells per drug.

The response-matching criterion is an energy-based welfare regret. Each candidate’s welfare is the aggregate over query states s of its negative energy distance to that state, $U(d) = \text{agg}_s (- \mathcal{E}(P_d, Q_s))$ with the aggregator taken as the minimum in the worst-case form, and the regret of the selected candidate $\hat{d}$ is

$$\text{Regret}(\hat{d}) = \max_{d \in \mathcal{L}} U(d) - U(\hat{d}) \geq 0,$$

where $\mathcal{L}$ is the observed library, so the reference is the best candidate actually available rather than an unobservable optimum. Minority-state coverage is reported on the same rankings as the cosine between

the candidate’s mean response and the mean response of the query’s minority state, rescaled from $[-1, 1]$ to $[0, 1]$. Regret is zero when the top-ranked candidate is optimal under that objective and increases as the selected candidate becomes less favourable. We report regret reduction relative to mean-signature retrieval, such that a positive value indicates lower regret under this response-matching criterion. This evaluation is intentionally closely aligned with the population-level information used by the retriever and is therefore interpreted as evidence of response fidelity rather than as an independent functional outcome.

Task-proximal biological evaluation. The same fixed rankings were next evaluated using mechanism-of-action (MoA) recovery and minority-state coverage. MoA recovery was quantified by normalized discounted cumulative gain (nDCG)³⁸, which rewards candidates sharing the query perturbation’s annotated mechanism when they occur near the top of the ranked list. Mechanism recovery is scored on the 600 leave-drug-out queries, that is on the 765 partial-observation queries less the 120 leave-MoA-out and 45 partial-library queries, whose candidate libraries are built without a member of the query mechanism class and for which recovery is therefore undefined. Selection is by split type. Within the 600, 21 queries happen to hold no mechanism-class member either and score exactly zero under every method, so they enter the comparison as ties rather than as a difference; dropping them leaves the median difference at 0.000 and moves the mean from -0.024 to -0.025 and the decreased and increased shares from 40.8% and 35.7% to 42.3% and 37.0%.

Minority-state coverage asks whether the top-ranked candidate better preserves the response of the less represented cellular state in the controlled heterogeneous-population construction. As with MoA recovery, this criterion is applied to rankings that have already been generated; the evaluator does not alter the query, the retrieval score or the candidate ordering. Differences between population-level and mean-signature retrieval are therefore attributable to how the same rankings score under the biological criterion rather than to refitting or re-ranking.

Cross-setting projection. To examine whether the minority-state result extends beyond the controlled partial-observation setting, we evaluated the top-ranked population-level and mean-signature candidates across 239 retrieval task instances spanning natural and constructed settings. These comprised 90 constructed cross-cell-line SciPlex3 tasks, 90 within-cell-line SciPlex3 tasks, 23 Frangieh tasks, 12 CD34+ tasks and 24 tasks using predicted candidate populations. For each task we report the difference in minority-state coverage between the population-level and mean-signature selections.

The 239 tasks are treated as a distribution of task-level differences rather than as independent biological replicates of a single experiment. In particular, the constructed SciPlex3 tasks share data source, preprocessing and candidate structure, and the three SciPlex3 cell lines constitute leave-one-cell-line-out variation within one experiment rather than leave-one-dataset-out generalization. Thresholded counts of “distributionally dominant” tasks are therefore not used as a primary result; the continuous difference in coverage is reported instead.

External functional evaluation using GDSC2

To test whether transcriptomic retrieval rankings capture information that extends beyond the single-cell expression measurements used to construct them, we evaluated the SciPlex3 rankings against drug-response profiles from the Genomics of Drug Sensitivity in Cancer^{28,29}, using the GDSC2 screening set (release 8.5; Data availability). The external criterion was defined as similarity between the functional

response profiles of two drugs rather than as absolute drug potency, because the retrieval task asks which candidate produces a response most similar to the query rather than which candidate is most cytotoxic.

SciPlex3 compounds were matched to GDSC2 by normalized drug name, with substring matching used only as a fallback. After removal of one duplicate mapping and restriction to the GDSC2 screening platform, 35 compounds remained. GDSC1 and GDSC2 measurements were not pooled because they derive from different assay and dose-range settings. For each matched compound, its functional profile was the vector of fitted dose-response AUC values across GDSC2 cell lines. The three cell lines used for the transcriptomic retrieval experiments, A549, K562 and MCF7, were excluded from these profiles, leaving 966 cell lines, so the functional evaluation was performed in biological contexts not used to construct the RNA rankings.

Cell lines differ systematically in their overall sensitivity to drug treatment, which can induce positive correlations between otherwise unrelated compound profiles. We therefore removed this shared cell-line effect before calculating drug-drug similarity. For cell line l and compound d , the centred response was

$$\widetilde{\text{AUC}}_{d,l} = \text{AUC}_{d,l} - \frac{1}{|\mathcal{D}_l|} \sum_{d' \in \mathcal{D}_l} \text{AUC}_{d',l},$$

where $\mathcal{D}_l$ contains all GDSC2 compounds measured in cell line l , rather than only the 35 compounds overlapping SciPlex3. Functional similarity between query drug q and candidate drug c was then defined as

$$F(q, c) = \rho_s \left(\widetilde{\text{AUC}}_{q,\bullet}, \widetilde{\text{AUC}}_{c,\bullet} \right),$$

where ρ_s denotes Spearman correlation across cell lines shared by the two compounds. Drug pairs measured in fewer than 100 common cell lines were excluded. This yielded 591 usable drug pairs, with a median of 909 shared cell lines per pair. The corresponding uncentred functional similarity was retained as a sensitivity analysis.

For each SciPlex3 cell line, retrieval was evaluated in a leave-one-drug-out design. The 10 μM response population of the held-out compound served as the query, and the remaining matched compounds formed the candidate library, giving 34–35 candidates per query. Each RNA-based retrieval rule produced a candidate ranking without access to GDSC2 data. Agreement between that ranking and the external functional ordering $F(q, c)$ was quantified by Spearman correlation across candidates.

The primary RNA-derived rankings were energy-distance population retrieval and control-referenced mean-cosine retrieval. To test whether apparent functional agreement required cell-to-cell distributional information, we additionally used a response-magnitude matching score,

$$M(q, c) = - \left| \|\vec{d}_q\| - \|\vec{d}_c\| \right|,$$

which prefers candidates whose overall transcriptional response magnitude is similar to that of the query but uses no within-population structure. A query-independent magnitude score $\|\vec{d}_c\|$ was retained as a diagnostic control. An uncorrected mean-cosine score based on treated means without matched-control subtraction and a potency-matching diagnostic were analysed as additional sensitivity checks rather than as primary retrieval baselines.

To estimate the component of the energy-based association not explained by response magnitude, we computed a partial rank correlation with the magnitude-matching channel removed. Retrieval scores and functional similarities were rank-transformed, each was residualized with respect to $M(q, c)$, and

the correlation between the two residuals was reported as the magnitude-adjusted association.

Per-query Spearman correlations are summarized by their median within each SciPlex3 cell line. Queries from the same cell line share the same candidate library and are therefore not treated as independent biological replicates. The three cell lines provide the higher-level analysis unit; query-level paired tests are reported only as descriptive analyses where indicated.

The comparison between energy retrieval and the response-magnitude scalar is pre-specified as a pooled test across the three SciPlex3 cell lines; per-line ratios are given in Supplementary Methods.

Protein-based evaluator-shape experiment

Frangieh Perturb-CITE-seq³⁰ provides matched RNA and surface-protein measurements from the same cells, allowing the statistical form of an external evaluator to be changed while holding its biological measurement fixed. RNA and protein matrices were joined by cell barcode before analysis; row position was not used because the released modality-specific matrices differ in cell number and ordering. Of the 24 measured antibodies, four isotype controls were removed after normalization, leaving 20 surface-protein features. Antibody-derived tag counts³⁹ were centred-log-ratio normalized within each cell, $y_{ij} = \ln(1 + x_{ij}) - m^{-1} \sum_k \ln(1 + x_{ik})$ for cell i and antibody j , with a pseudocount of one, a natural logarithm and the per-cell mean taken over all $m = 24$ measured antibodies including zero counts and the four isotype controls; normalization therefore precedes isotype removal. This is the centred log-ratio of the $(1 + x)$ vector and differs from implementations that exclude zero entries from the geometric mean.

All analyses were performed separately within the Control, IFN γ and Co-culture conditions to avoid mixing immune context with perturbation identity. Knockouts represented by at least 60 barcode-matched cells were retained, and populations were subsampled to at most 250 cells using a fixed random seed.

Protein-based evaluators. For knockout k in immune condition c , let $\bar{d}_{k,c}^{\text{prot}}$ denote its mean control-referenced protein response and let $D_{k,c}^{\text{prot}}$ denote the corresponding set of per-cell protein responses. The mean-based evaluator was defined as

$$E_{\text{mean}}(q, d) = \cos \left(\bar{d}_{q,c}^{\text{prot}}, \bar{d}_{d,c}^{\text{prot}} \right).$$

The population-based evaluator was defined from the same cells and protein measurements as

$$E_{\text{pop}}(q, d) = -\mathcal{E} \left(D_{q,c}^{\text{prot}}, D_{d,c}^{\text{prot}} \right),$$

where $\mathcal{E}$ is the energy distance. The negative sign converts distance to similarity so that larger values indicate a closer match under both evaluators.

For both definitions, protein responses were referenced to the matched control population from the same immune condition. Thus,

$$\bar{d}_{k,c}^{\text{prot}} = \text{mean} \left(X_{k,c}^{\text{prot}} \right) - \mu_{0,c}^{\text{prot}},$$

and

$$D_{k,c}^{\text{prot}} = \left\{ x_{i,k,c}^{\text{prot}} - \mu_{0,c}^{\text{prot}} \right\}_{i=1}^{n_{k,c}}.$$

Using the same matched control for the two evaluator forms ensures that their difference arises from whether protein responses are summarized by their mean or retained as a cellular population, rather than from a change in biological reference.

Fixed RNA-derived rankings. The RNA-derived rankings were computed once and then evaluated against both protein criteria. The primary rankings were energy-distance population retrieval and control-referenced mean-cosine retrieval. The query populations, candidate libraries and RNA-derived candidate scores were identical under the two protein evaluators, and no protein measurement was available to any retrieval method. The experiment therefore changes only the statistical representation used to judge the same RNA-derived ranking.

Response-magnitude control. Because a population distance can also encode differences in overall response magnitude, we included a query-dependent scalar control that uses no cell-to-cell distributional information. For query q and candidate d , magnitude matching was defined as

$$M(q, d) = - \left| \left| \bar{d}_{q,c}^{\text{RNA}} \right| \right| - \left| \left| \bar{d}_{d,c}^{\text{RNA}} \right| \right|.$$

This ranking favours candidates with a transcriptional response magnitude similar to that of the query while discarding all within-population structure. The same magnitude ranking was evaluated against both protein criteria.

For each query, agreement between an RNA-derived ranking and a protein-based evaluator was quantified by Spearman correlation across that query’s candidate knockouts. The statistic is therefore a rank correlation over candidates, computed once per query, and never a correlation over pooled query–candidate pairs. The headline value for each ranking and each evaluator form is the median of those per-query correlations over all 659 queries. The same medians were also taken within each of the three immune conditions, which are reported separately and treated as the higher-level analysis unit; queries within a condition share candidate structure and were not interpreted as independent biological replicates.

Natural heterogeneity and large-scale validation

We next examined the information requirements for population-level retrieval in datasets in which cellular heterogeneity was not created by mixing experimentally distinct source populations. Patient-derived glioblastoma was used to test the requirements in naturally heterogeneous tissue, and Tahoe-100M was used to assess the same quantities at substantially larger scale. The glioblastoma cohort was not used as a retrieval benchmark; Tahoe-100M additionally served as the substrate for the source-to-target intervention-identification benchmark described below.

Patient-derived glioblastoma. The ZhaoSims2021 dataset¹⁶ contains acute-slice cultures and surgical biopsies from ten glioblastoma patients exposed to six compounds: Ana-12, Ispenisib, RO4929097, Tazemetostat, etoposide and panobinostat. Expression counts were normalized to 10^4 per cell, log1p-transformed and reduced to 2,000 highly variable genes. Because only one patient received more than two compounds, the cohort does not support a sufficiently replicated leave-one-drug-out retrieval benchmark. We therefore use it only to measure state-specific response, recoverability and the extent to which population-level similarity differs from compartment-specific response ordering.

675 **Cellular-compartment assignment.** Cells were assigned to malignant and myeloid compartments using canonical marker scores; ambiguous cells were excluded. Drug responses were defined relative to patient- and compartment-matched controls. Assignment criteria, retained-cell counts and threshold-sensitivity analyses are reported in Supplementary Note 3.

679 **Differential response in natural tissue.** State-specific perturbation responses were measured between malignant and myeloid compartments. For patient p , drug d and compartment k , the response vector was

$$r_{p,d,k} = \text{mean} \left(X_{p,d,k}^{\text{treated}} \right) - \text{mean} \left(X_{p,k}^{\text{control}} \right).$$

682 Only patient–drug combinations with at least 50 treated cells in both compartments were included. Controls were matched within both patient and compartment. Differential response was summarized as

$$R_{p,d} = \cos \left(r_{p,d,\text{malignant}}, r_{p,d,\text{myeloid}} \right).$$

684 This construction removes pre-existing differences between patients and cellular compartments before comparing the directions of their drug-induced responses.

686 **Recoverability in natural tissue.** Response-identity recoverability was assessed from within-patient drug-pair splits by comparing unsupervised partitions with a supervised ceiling. Patient-level bootstrap and method-pair sensitivity analyses are reported in Supplementary Note 3.

689 **Proxy for downstream decision relevance.** The glioblastoma cohort does not contain enough drug coverage to test whether population-level retrieval directly changes the preferred candidate. We therefore quantified a more limited property: whether drug similarity at the whole-population level preserves the ordering observed within a specific cellular compartment.

693 For each pair of drugs measured in the same patient, we first computed the cosine similarity between their population-mean response signatures and the cosine similarity between their malignant-compartment response vectors. Agreement between these two similarity structures was summarized by Spearman correlation across drug pairs.

697 That whole-population comparison gives a Spearman correlation of +0.878, but the population-mean signature is the equal-weight average of the two compartment responses, so the malignant compartment contributes half of it and part of the agreement is arithmetic. We therefore additionally used a disjoint comparison. For drug pair (a, b) , myeloid-response similarity was defined as

$$S_{a,b}^{\text{myeloid}} = \cos \left(r_{a,\text{myeloid}}, r_{b,\text{myeloid}} \right),$$

701 and malignant-response similarity as

$$S_{a,b}^{\text{malignant}} = \cos \left(r_{a,\text{malignant}}, r_{b,\text{malignant}} \right).$$

702 We then computed

$$\rho_{\text{disjoint}} = \rho_S \left(S_{a,b}^{\text{myeloid}}, S_{a,b}^{\text{malignant}} \right),$$

703 using only drug pairs available within the same patient. The two sides of this comparison contain no
704 shared cells and each compartment is referenced to its own matched control. This statistic is interpreted
705 as a proxy for the amount of candidate-ordering headroom available to population structure, not as a
706 direct measure of retrieval performance. No pass-fail threshold was applied.

707 **Large-scale validation in Tahoe-100M.** Tahoe-100M plate 3 was used to test the same requirements at
708 larger scale. We evaluated two independently defined states, released cell-cycle labels and a control-
709 derived k -means partition, and quantified state-specific response alignment, response-identity recover-
710 ability and agreement of drug ordering across disjoint states. Dataset filtering and all state-definition
711 details are reported in Supplementary Note 3.

712 For each eligible drug-cell-line condition, the response of state k was defined relative to its own
713 state-matched controls,

$$r_{d,c,k} = \text{mean} \left(X_{d,c,k}^{\text{treated}} \right) - \text{mean} \left(X_{c,k}^{\text{control}} \right).$$

714 Differential response was measured as the cosine between the two state-specific response vectors,

$$R_{d,c} = \cos \left(r_{d,c,0}, r_{d,c,1} \right).$$

715 A condition was included only when at least 50 cells were available in each required treated and control
716 state.

717 Finally, a disjoint state-ordering statistic was used as a large-scale proxy for downstream decision
718 relevance. Within each cell line, cosine similarity was computed for every eligible pair of drug responses
719 separately in the two non-overlapping cellular states. Spearman correlation between the two resulting
720 drug-drug similarity matrices was then calculated,

$$\rho_c = \rho_S \left(S_{a,b}^{(0)}, S_{a,b}^{(1)} \right).$$

721 High correlation indicates that the two cellular states preserve similar relative ordering among drug
722 responses, whereas lower correlation leaves greater scope for state-specific information to change candi-
723 date preference. As in the glioblastoma analysis, this quantity does not directly test whether population-
724 level retrieval improves a drug ranking.

725 These are intrinsic-state diagnostics and are reported separately from the intervention-identification
726 benchmark below, which uses the same plate. Detailed recoverability methods are in Supplementary
727 Note 3.

728 **Source-to-target intervention identification on Tahoe-100M**

729 The retrieval experiments above rank candidates whose responses in the query context are observed.
730 An intervention task instead asks which compound will carry a starting population into a desired one,
731 so the candidate’s effect in that context is unknown. The scope of that question here is narrower than
732 intervention design, and the naming follows the scope. The desired population is a real treated popu-
733 lation generated by one compound of the library, and the outcome is the rank that compound receives,
734 so the task measures whether population information helps identify the intervention that produced an
735 observed response. It does not measure therapeutic utility, and no target state is specified independently
736 of a compound that already reaches it. We defined this task on Tahoe-100M plate 3.

Task definition and eligibility. A context is one cell line. For a held-out context c and a query drug q , the retriever receives the vehicle (DMSO) cells of c and an observed treated population $X_{q,c}$, and ranks all 92 candidate compounds; it never sees q . A condition entered the grid if it had at least 100 treated and 200 vehicle cells, which admitted 3,992 eligible queries over 44 usable cell lines from the 4,600 conditions present on the plate. Each query was evaluated at five fixed seeds (13, 29, 47, 71, 101), giving 19,960 query-seed pairs. Source and target populations were drawn without replacement at up to 200 cells each, all available cells being taken where a condition held fewer, and responses were referred to the source mean, $P = \{x_i - \bar{x}_{\text{source}}\}$.

Oracle arm. In the oracle arm every candidate’s response in the held-out context is taken from the observed data, which removes the forward model from the loop and bounds what any predictor could deliver. Cell disjointness is enforced through the library preparation: the target population is drawn from one sublibrary half and every candidate bank from the other, at up to 100 cells per candidate, all cells in that half being taken where a candidate has fewer, without replacement, so the generating drug is drawn from the same half as its 91 competitors and is not penalized by a technical shift. This arm is an information ceiling and not a deployment result.

Prediction arm. In the prediction arm no treated cell of the held-out context enters any fit or any ranker. Candidate responses are produced by a forward predictor fitted on the other 43 cell lines. Six predictors were evaluated. Three are additive: they emit one signature per drug and apply it identically to every source cell, so the candidate population is the vehicle population rigidly translated. These are an average effect δ_d pooled over training contexts; a nearest-context transfer taking δ_d from the training context whose vehicle mean is closest in cosine; and a linear-latent variant truncating the average effect to the principal components carrying 80% of training variance. Three are not additive. The *state-conditioned* predictor estimates the same average effect separately for each of the three cell-cycle states released with the plate, G1, S and G2M, and applies it to each source cell according to its own state, which costs one extra grouping and no new hyperparameter. Each per-state effect is averaged over the training contexts in which that state holds at least 20 cells; groups below the floor are dropped from the average rather than estimated, and 359 of the 12,275 cell-line, drug and state groups fall below it, 294 of them in S phase. A drug left with no usable estimate for a state would keep its pooled effect there and be additive for that state alone; this did not occur for any of the $44 \times 92 \times 3$ signature slots, so the predictor is non-additive over its whole output. The *Gaussian optimal-transport* map is a Bures map fitted in a 50-component training principal-component basis, from the pooled within-context vehicle covariance to the drug’s pooled within-condition treated covariance, with the mean shifted by the average latent effect and applied to the held-out vehicle cells as a latent edit so that the part of each cell outside the subspace is carried through; every cloud is centred on its own condition mean before pooling, so a covariance here is within-condition cell scatter. *Nearest context, cells* transfers the neighbouring context’s actual treated cells for the drug, shifted by the difference of the two vehicle means; it is the cheapest predictor in this set that can carry a non-additive response shape, and it is included because a set of purely additive predictors would answer the question by construction rather than by measurement. Each principal-component basis is fitted on up to 50,000 cells sampled from the training contexts, and each context-drug fit uses up to 200 cells. The additive three generate no state-specific differential response, which is an algebraic property of the model class and not a training failure.

Scoring and evaluation. Candidates were scored by the three rules of Fig. 1a: a direction-only mean cosine, a magnitude-aware mean L_2 , and the population route, primarily the unbiased U-statistic energy distance, with MMD and sliced Wasserstein as secondary variants. The secondary scores use settings specific to this task: sliced Wasserstein over 200 random projections at 100 quantiles, and a single RBF bandwidth fitted once by the median heuristic on the candidate library and held fixed across queries, rather than the 128-projection and pooled multi-scale settings of the observed-population arm. Mean reciprocal rank of the true intervention is the primary outcome; Hit@1, Hit@5 and Hit@10 are secondary. Intervals are 2,000-fold bootstrap intervals resampled over cell-line clusters, which are the independent units of this design. A decision is counted as corrected when the population route places the true drug first and the reference does not, and as broken in the reverse case.

788 Headroom, response detectability and the conditional bottleneck analysis

Headroom and its null. Headroom is defined per query as $H = 1 - \text{RR}_{\text{ref}}$, the room the direction-only mean route leaves for any other scorer to improve on, and the gain is $G = \text{RR}_{\text{pop}} - \text{RR}_{\text{ref}}$ on the same query. A Spearman correlation between H and G is partly arithmetic, because the two share RR_{ref} with opposite signs and reciprocal rank is capped at 1. The null quantifies exactly that. Within each cell line, the population route’s reciprocal ranks are randomly permuted across queries, which destroys any pairing between the two scorers while preserving both marginal distributions, the $\text{RR} \leq 1$ ceiling and the cell-line clustering; G is then recomputed against the unpermuted reference and the correlation retaken. Two hundred permutations at a fixed seed give the interval reported in Fig. 5e.

Response detectability. For each held-out cell line c and query drug q , equal numbers of cells treated with q and vehicle cells of c form a balanced two-class problem, at 300 per arm or at the size of the smaller pool where that is below 300, sampled at a fixed seed, so chance is 0.5 by construction. The supervised ceiling A_{sup} is the out-of-fold accuracy of a five-fold cross-validated logistic probe ($C = 1$) on those cells. The unsupervised value A_{unsup} is k -means with $k = 2$ on the same cells, scored by best-permutation accuracy, so cluster-to-label matching is free and the unsupervised arm is not penalized for arbitrary label order. The labels are external, being which well a cell came from, rather than an expression-derived cell state. The supervised arm is a ceiling on the information present in the representation.

Conditional bottleneck regression. Regressing the gain G directly on candidate explanators inherits the same arithmetic as the headroom correlation. The specification reported in Fig. 5h therefore takes the population route’s own reciprocal rank as the outcome and includes the mean route’s reciprocal rank as a covariate, so that a coefficient answers a conditional question: given how well the mean route did on this query, does this explainer predict how well the population route did. The covariate is a control and is not plotted. The explanators are the interaction share, its cross-fitted reproducibility $R_{\text{int}} = \cos(I_A, I_B)$, response detectability A_{sup} , and, in a separate model reported for contrast, the naive response cosine $1 - \cos(r_1, r_2)$. Estimation is ordinary least squares with cluster-robust standard errors at the cell line, which is the independent unit of the design, and all predictors are standardised to zero mean and unit variance so that coefficients are comparable across explanators on one axis. Queries missing any explainer are dropped listwise, which is why the reported n of 3,618 is below the 3,992 of the retrieval arm.

817 Cross-fitted drug-by-state interaction

818 Testing whether a drug’s effect depends on cell state requires a statistic that measures interaction rather
 819 than effect size. Writing r_s for the state- s response, the difference between mean treated and mean vehicle
 820 cells of that state, with states 1 and 2 the G1 and G2M cell-cycle calls released with the plate, the quantity
 821 of interest is the interaction vector $I = r_1 - r_2$. Its squared norm cannot be used directly: with finite arms,
 822 sampling noise alone makes $\|I\|$ large, and the same mechanism inflates $1 - \cos(r_1, r_2)$ for the weakest
 823 drugs.

824 Each of the four arms is therefore split into disjoint halves A and B , the interaction is estimated once
 825 in each, and the two estimates are multiplied:

$$S_{\text{int}} = \frac{\langle I_A, I_B \rangle}{p}, \quad S_{\text{main}} = \frac{\langle M_A, M_B \rangle}{p}, \quad M = \frac{1}{2}(r_1 + r_2),$$

826 with p the number of genes. Under the additive null $E[I] = 0$ and the two halves are independent,
 827 so $E[S_{\text{int}}] = 0$ exactly: noise makes I_A and I_B point in unrelated directions and their inner product
 828 averages to zero, whereas a real state-dependent effect survives because both halves estimate the same
 829 non-zero I . The vehicle arms are split as well; sharing one vehicle mean between halves would place the
 830 same control noise in I_A and I_B and correlate them under the null. Splits are library-preparation-disjoint
 831 wherever a condition spans at least two sublibraries, and a plain random half of the cells otherwise.
 832 Arms were capped at 400 cells and required at least 50, and ten A/B splits were averaged per query.

833 Two derived quantities are used. The *interaction share*, $S_{\text{int}}/(S_{\text{int}} + S_{\text{main}})$, is scale-free and has a
 834 cross-fitted numerator and denominator, so neither term is noise-inflated; dividing by a plain response
 835 norm would reintroduce exactly the single-sample denominator that sank the cosine statistic. $R_{\text{int}} =$
 836 $\cos(I_A, I_B)$ asks whether the interaction reproduces across halves. As a positive control, synthetic condi-
 837 tions were built from the plate’s own cells by taking the G1 cells from one drug and the G2M cells from
 838 another, so interaction is present by construction in material with the same states and arm sizes as the
 839 queries.

840 The cosine statistic $1 - \cos(r_1, r_2)$ is retained only as a descriptive sensitivity analysis and as the his-
 841 torical comparator in Fig. 5g. It is not used as a differential-response measure anywhere that a conclu-
 842 sion depends on it. Where earlier analyses in this study used the induced-response cosine to characterize
 843 predictors or tissue, that use is descriptive alignment between two response directions and is labelled as
 844 such.

845 Statistical analysis

846 Statistical units follow the experimental design rather than the number of cells or query–candidate com-
 847 parisons: cells estimate a perturbation-response population but are not independent replicates of re-
 848 trieval performance⁴⁰.

849 The partial-observation arm departs from this rule: each query is treated as the unit of analysis
 850 although the queries are not independent. Of the 765 queries, 720 hold out one of 143 drugs and the
 851 remaining 45 pool the unmeasured drugs of the partial-library split into a single batch query; the 600
 852 leave-drug-out queries on which mechanism recovery is scored come from 130 of those drugs. Each
 853 drug is scored across three shared cell lines, several library fractions and up to five random seeds, so the
 854 765 queries rest on 420 distinct drug-by-cell-line contexts. No cluster-level resampling was applied, so
 855 the P values and intervals reported for that arm are anticonservative with respect to the drug as a unit.

856 Paired comparisons between retrieval methods on the same queries used two-sided Wilcoxon signed-

rank tests; independent groups used Mann–Whitney U tests; monotonic associations used Spearman rank correlation. Across pre-specified strata, P values were adjusted by the Benjamini–Hochberg procedure⁴¹. Effect sizes are reported with every test; achieved power and the sample size required for 80% power are computed for the one stratum where they bear on the conclusion and are reported in Supplementary Note 1. Mechanism-of-action analyses were restricted to the 600 queries whose mechanism class remained represented in the candidate library; undefined queries were excluded rather than encoded as zero.

For the GDSC2 functional analysis, Spearman correlations were computed per query across its candidate drugs and summarized within each SciPlex3 cell line, which is the analysis unit; query-level paired tests are reported as nominal. The protein evaluator-shape experiment was summarized within the Control, IFN γ and Co-culture conditions, its three higher-level units. For HIR-Bench, cross-validation was grouped by the label-determining (α , conflict) pair, giving 28 effective parameter cells, and performance was summarized by pooled out-of-fold AUC. For the patient-derived glioblastoma analysis, uncertainty in the recoverability gap was estimated by cluster bootstrap at the patient level with 5,000 resamples, differences being taken within each drug-pair split before aggregation; four patients contributed evaluable splits. Intervention-retrieval intervals are 2,000-fold bootstrap resamples over cell-line clusters. Tahoe-100M summaries are taken across eligible drug–cell-line conditions or within-cell-line drug pairs according to the statistic, and both cellular-state definitions, released cell-cycle state and the control-derived k -means partition, were analysed in parallel.

Analyses were implemented in Python. Single-cell data handling used Scanpy⁴²; the k -means, Gaussian-mixture, principal-component and logistic-regression procedures named above are the scikit-learn implementations⁴³.

Data availability

All datasets analysed in this study are publicly available.

SciPlex3, Frangieh Perturb-CITE-seq and ZhaoSims2021 were obtained in harmonized form from scPerturb¹³, Zenodo record 13350497 ([doi:10.5281/zenodo.13350497](https://doi.org/10.5281/zenodo.13350497), release 1.4). The exact harmonized files used in the analyses were SrivatsanTrapnell2020_sciplex3.h5ad for SciPlex3, FrangiehIzar2021_RNA.h5ad and FrangiehIzar2021_protein.h5ad for Frangieh Perturb-CITE-seq, and ZhaoSims2021.h5ad for the patient-derived glioblastoma analysis. The primary SciPlex3 data are available from the Gene Expression Omnibus under accession GSE139944¹⁴; the original Frangieh and ZhaoSims2021 datasets are described in the corresponding publications^{16,30}. The CD34+ haematopoietic progenitor dataset is available from the Gene Expression Omnibus under accession GSE306429³².

Tahoe-100M¹⁵ was released by Vevo Therapeutics and the Arc Institute under CC0 and is distributed through the Arc Virtual Cell Atlas and Hugging Face (tahoebio/Tahoe-100M). This study uses plate 3 only, together with the released obs_metadata and sample_metadata tables. Analyses use the released preprocessed expression matrix comprising 4,158,278 cells and 2,304 highly variable genes in log1p-transformed space.

The external functional evaluation used GDSC2 fitted dose–response data from the Genomics of Drug Sensitivity in Cancer, release 8.5 (GDSC2_fitted_dose_response_270ct23.xlsx), available from <https://www.cancerrxgene.org>. The compound-matching table used to align the SciPlex3 and GDSC2 drug panels (drug_match_table.csv) is provided with the analysis code.

Processed analysis inputs and the scripts used to construct them from the source datasets are provided in the accompanying repository. Source-data tables underlying each figure panel are provided

900 with the corresponding analysis outputs.

901 **Code availability**

902 All code used for data processing, retrieval, evaluation and statistical analysis is available at <https://github.com/Boom5426/PopRetrieve>.
903

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906 **Author contributions**

907 **[[Author contributions, CRediT taxonomy: conceptualization, methodology, software, formal analy-**
908 **sis, investigation, data curation, writing (original draft), writing (review and editing), visualization,**
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910 **Competing interests**

911 **[[Competing interests statement. If none: "The authors declare no competing interests."]]**

912 **Additional information**

913 Supplementary Information is available for this paper: Supplementary Methods, Supplementary Notes
914 1–4 and Supplementary Tables 1–4.

915 Correspondence and requests for materials should be addressed to **[[Name]]**.

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